

Computer Graphics

COMP3421 and COMP9415

Never Stand Still

Unreal Engine (How to develop a 3D maze)

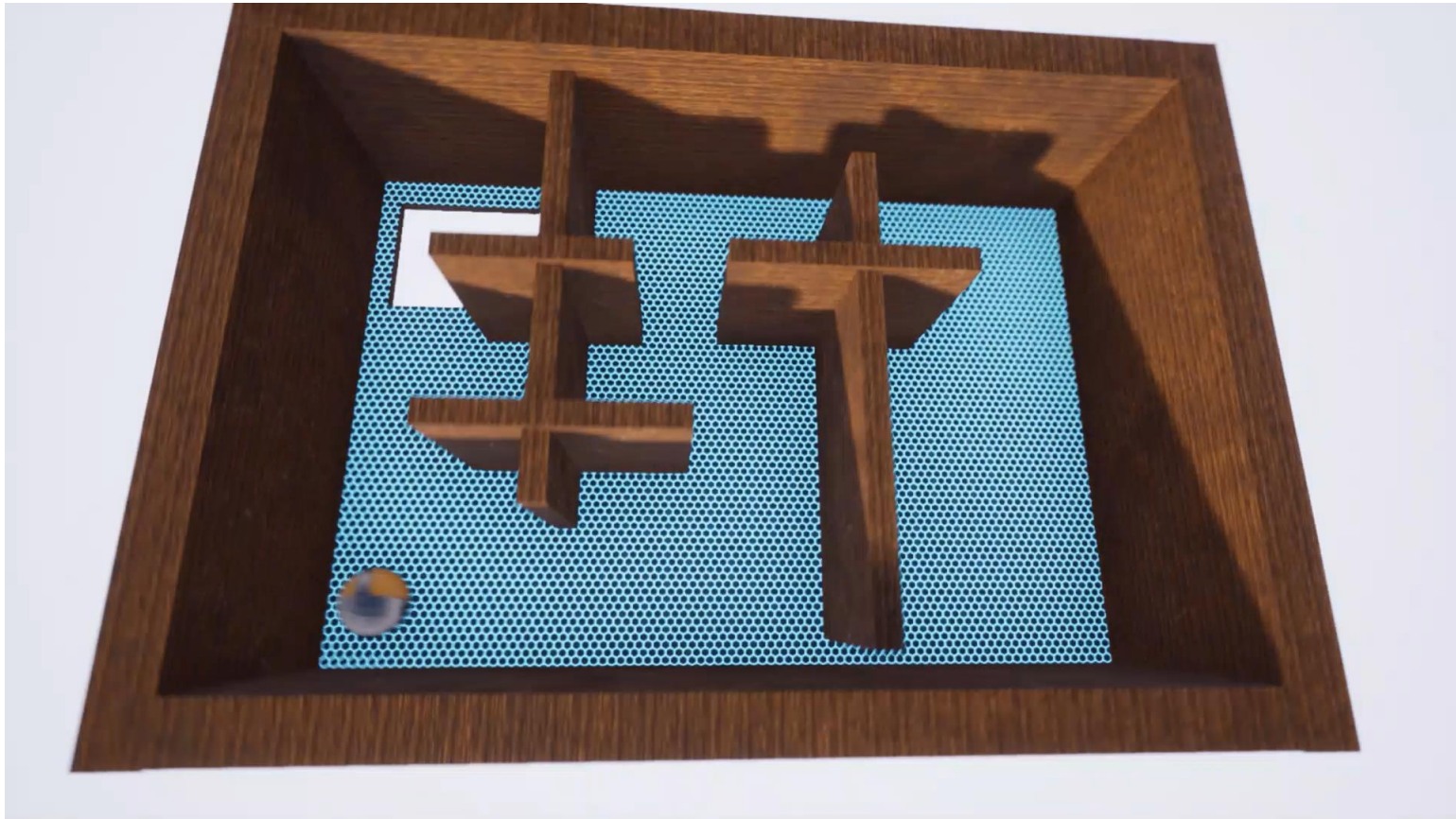
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School of Computer Science and Engineering
Term 3, 2023

Content overview

- Transforming the floor using keyboard and mouse input
- Add content to the level
- Add material to the objects
- Apply physics
- Geometry Brushes
- Static mesh and collision detection
- Create a simple win condition for levels
- Using trigger box
- Restarting and loading a level
- Publishing

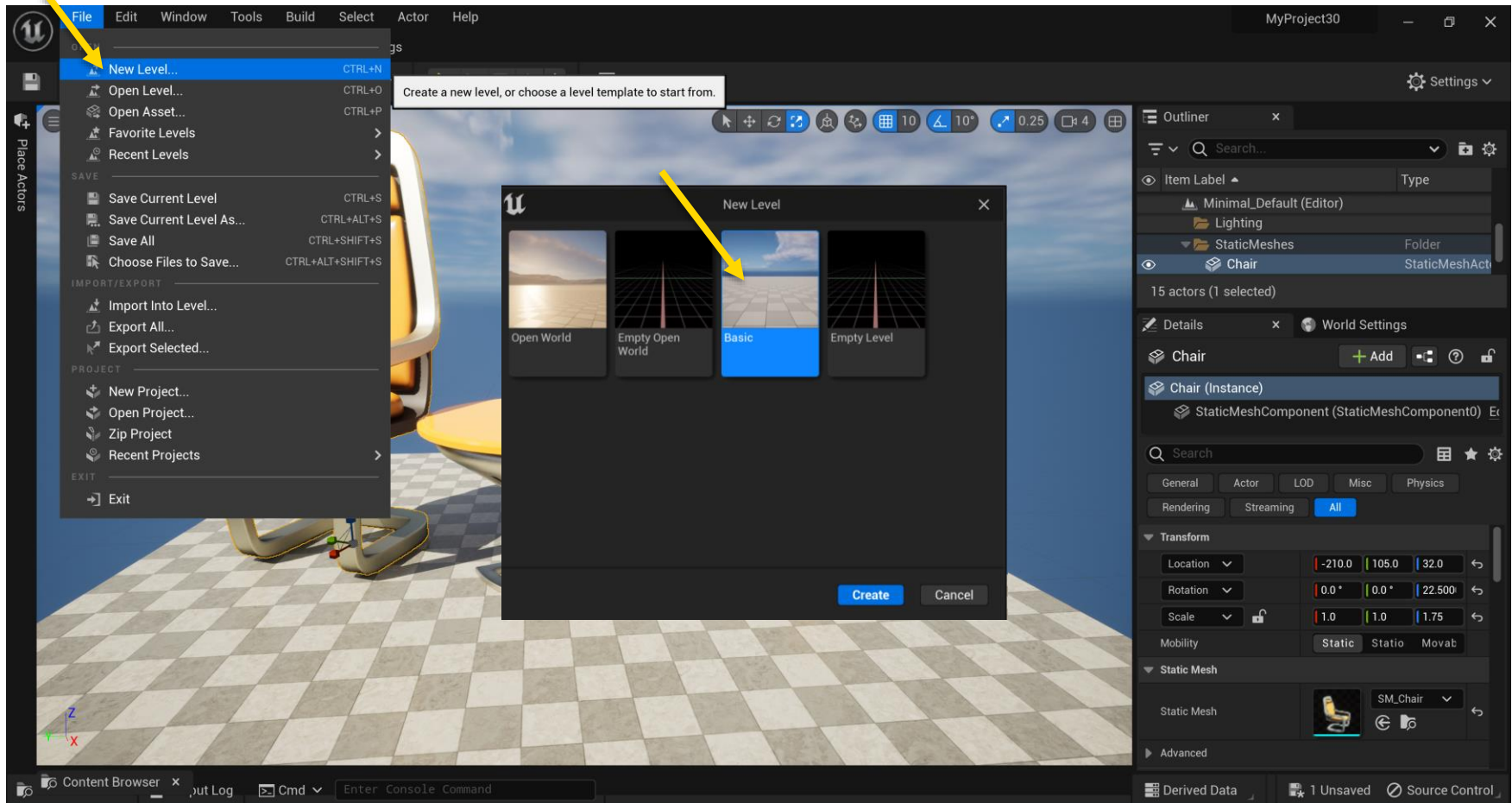
The first project of the course (Maze game)

- The project is a maze game.



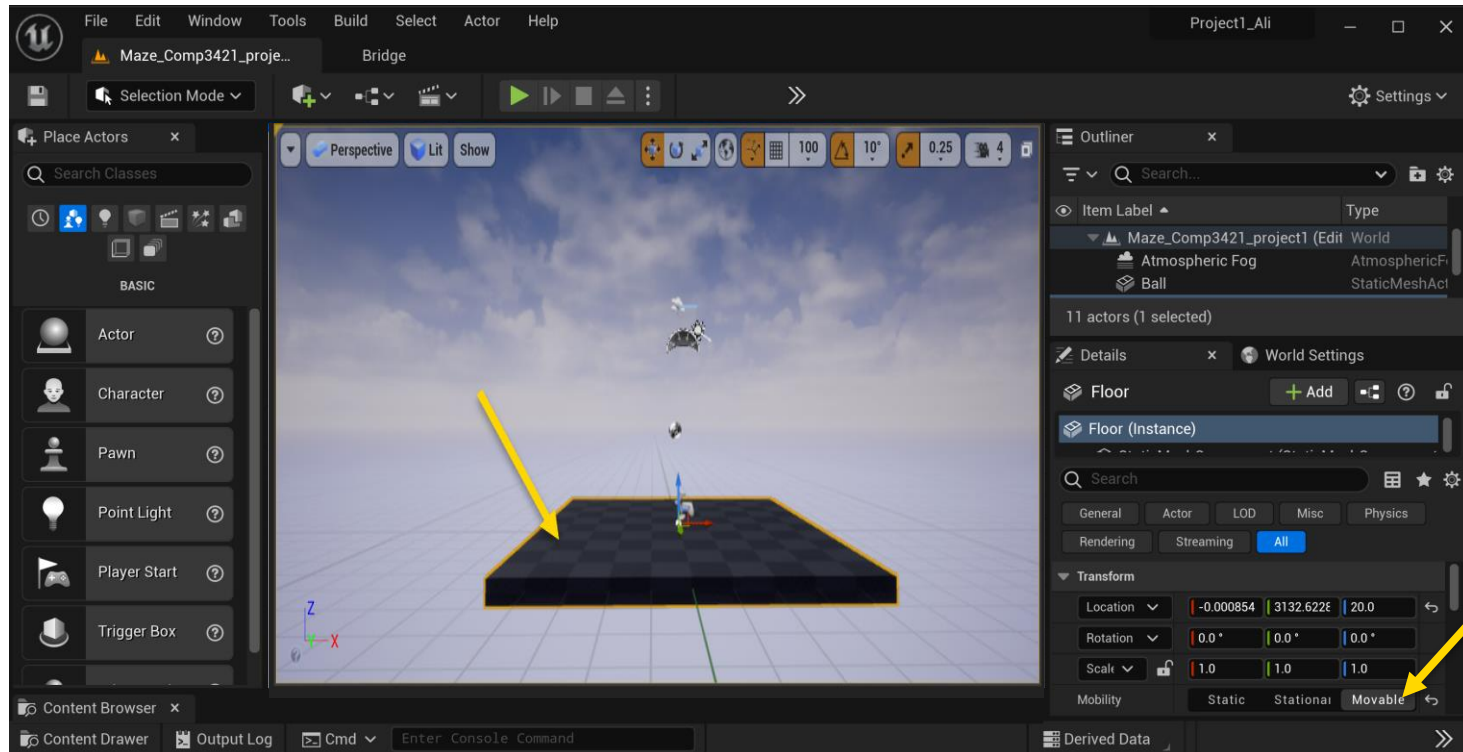
Create a new level

File menu > New Level > Basic



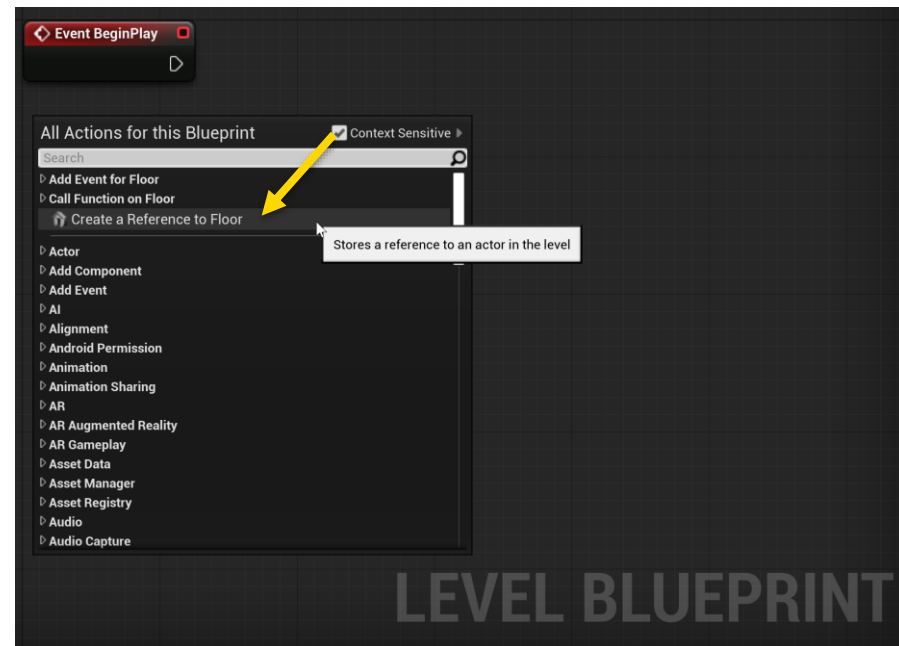
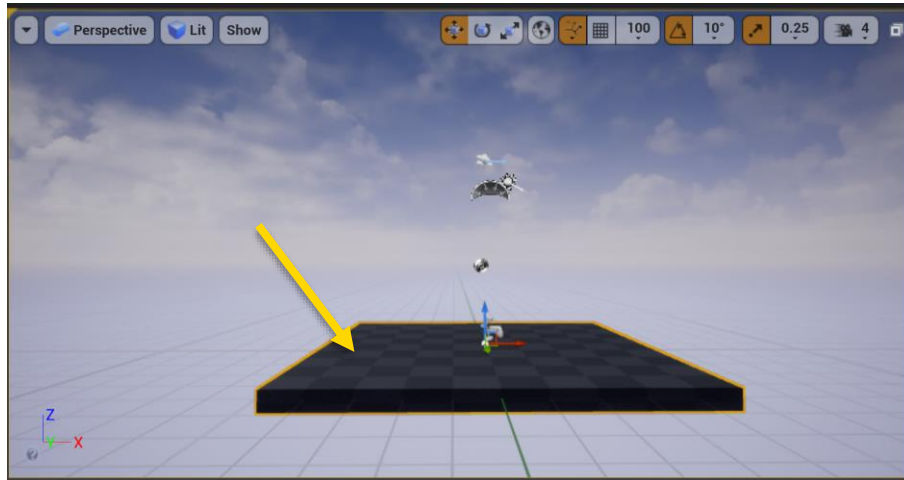
Make the floor movable

To enable users to tilt the floor to control the maze with the keyboard > Transform settings > Change the mobility from **static** to **movable**.



Tilt the floor using keyboard keys

- Select the floor in the viewport > Go to the blueprint editor > Click: “Create a reference to floor”.



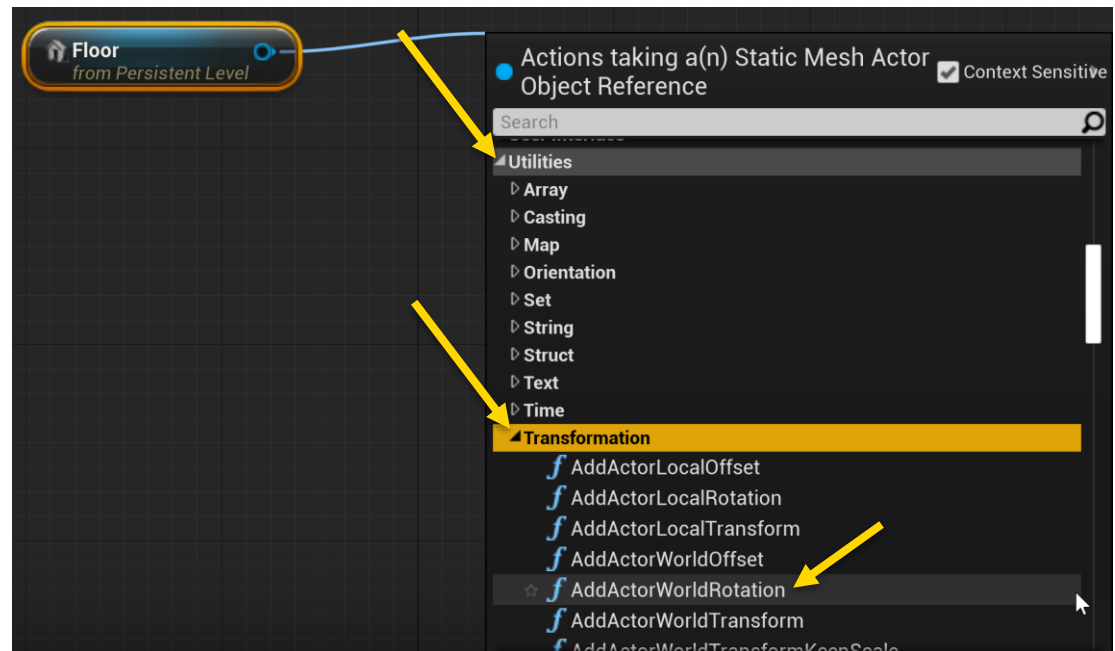
Tilt the floor using keyboard keys

- Hovering mouse over the pin, tells us the type of data that we are dealing with.
- For the floor, we are dealing with a static mesh actor object reference.



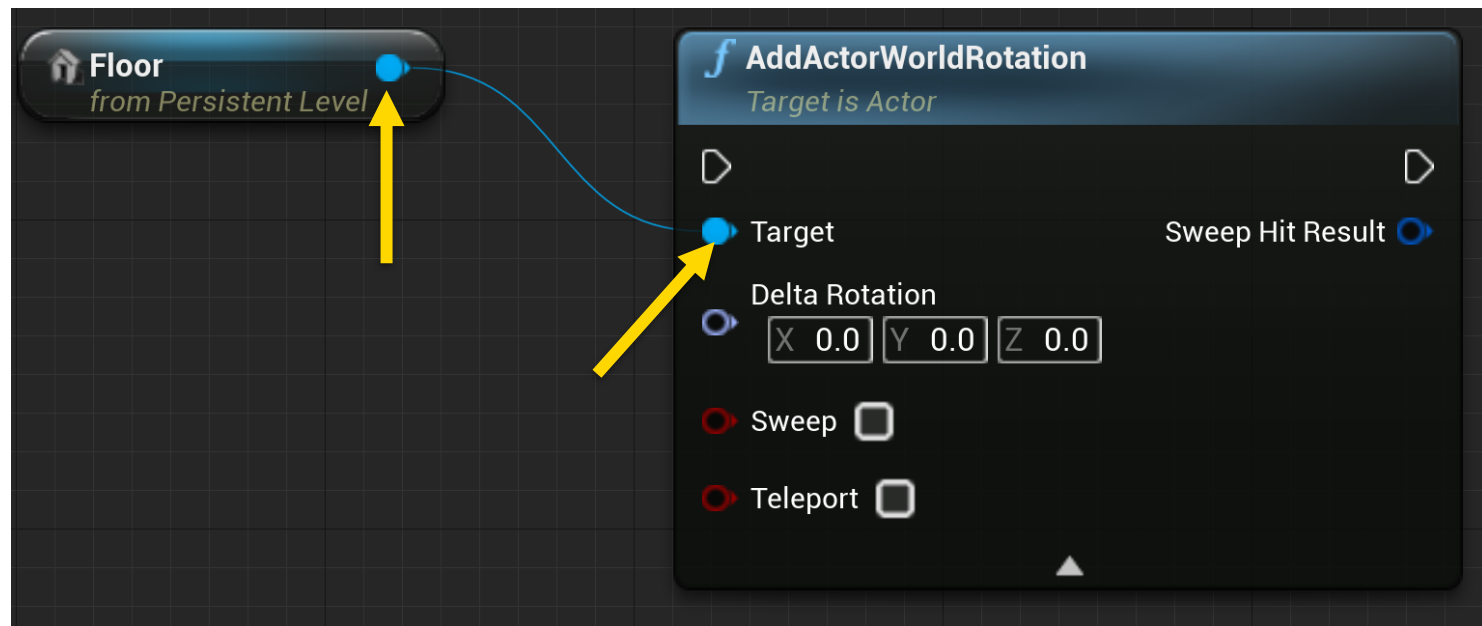
Tilt the floor using keyboard keys

- We need a function to enable us rotate the floor actor.
- A transform contains the actor's location, rotation and scale.
- Utilities category
 - > transformation
 - > **Add Actor World Rotation**



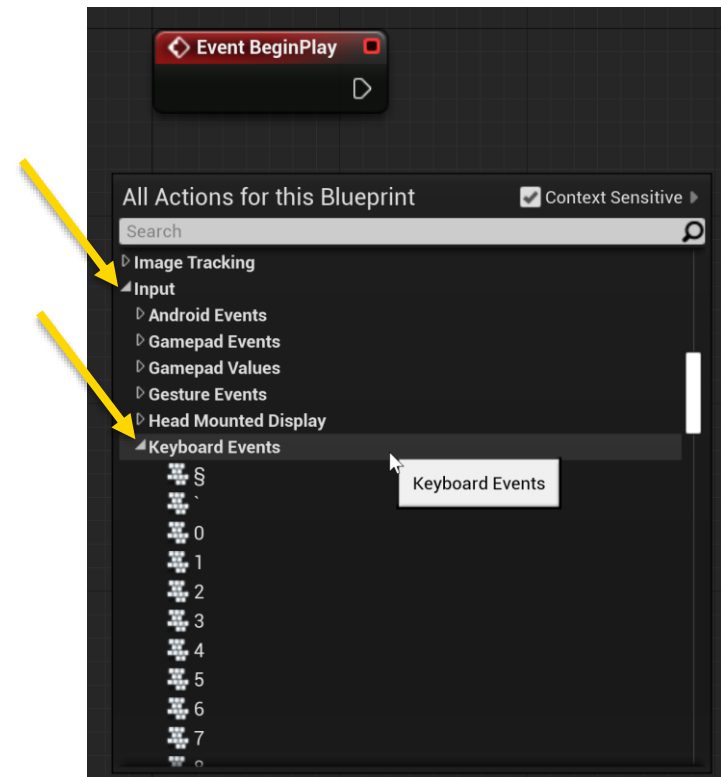
Tilt the floor using keyboard keys

- This function allows us to rotate our floor
- It has Delta rotation: roll, pitch and yaw.



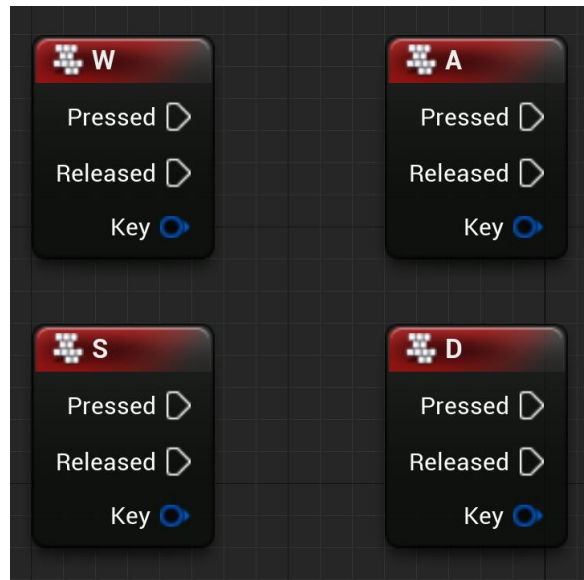
Tilt the floor using Input events

- Input events react to player input, such as when we press a key on the keyboard or mouse
- Go to the Input category > Keyboard Events



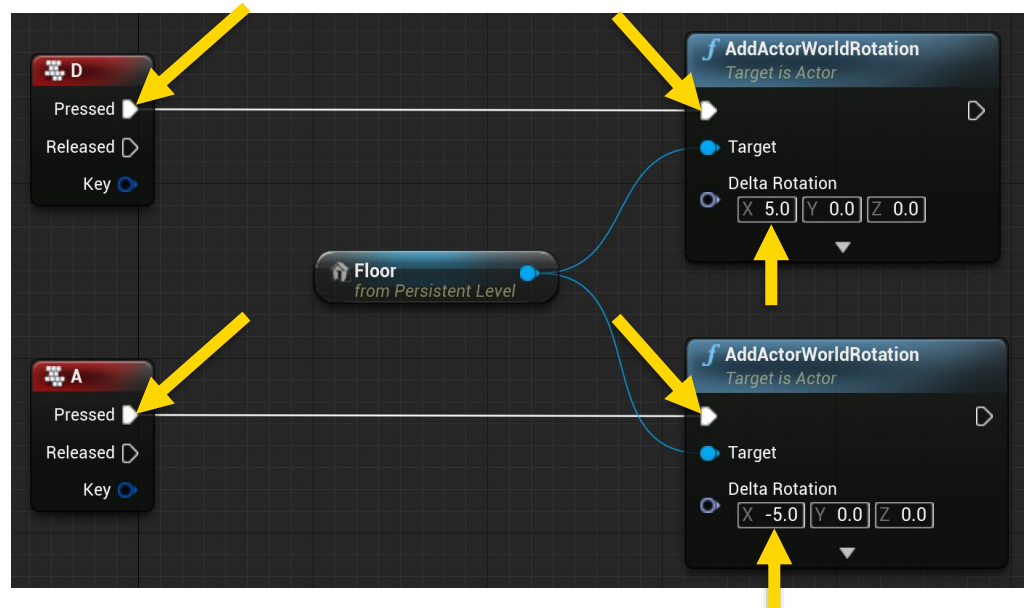
Tilt the floor using Input events

- Keyboard Events > W
- Keyboard Events > S
- Keyboard Events > A
- Keyboard Events > D



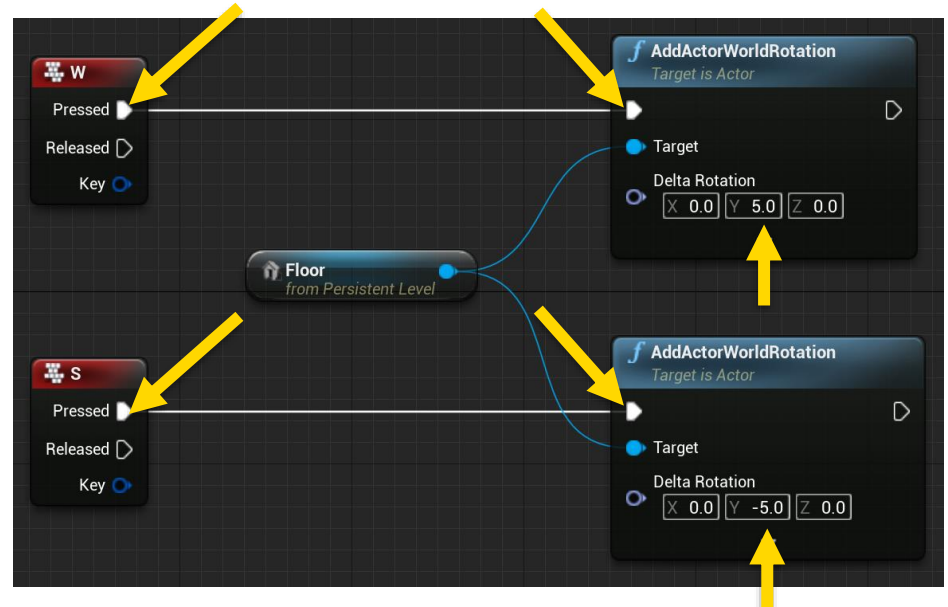
tilt the floor using keyboard keys

- The D key rotates the floor by 5 units to the right and the A key rotates the floor by 5 units to the left (Use the X axis).
- Add two 'Add actor world rotation' to the floor actor.
- Connect the input nodes (D & S) Pressed pin to the 'Add Actor World Rotation' input pin.



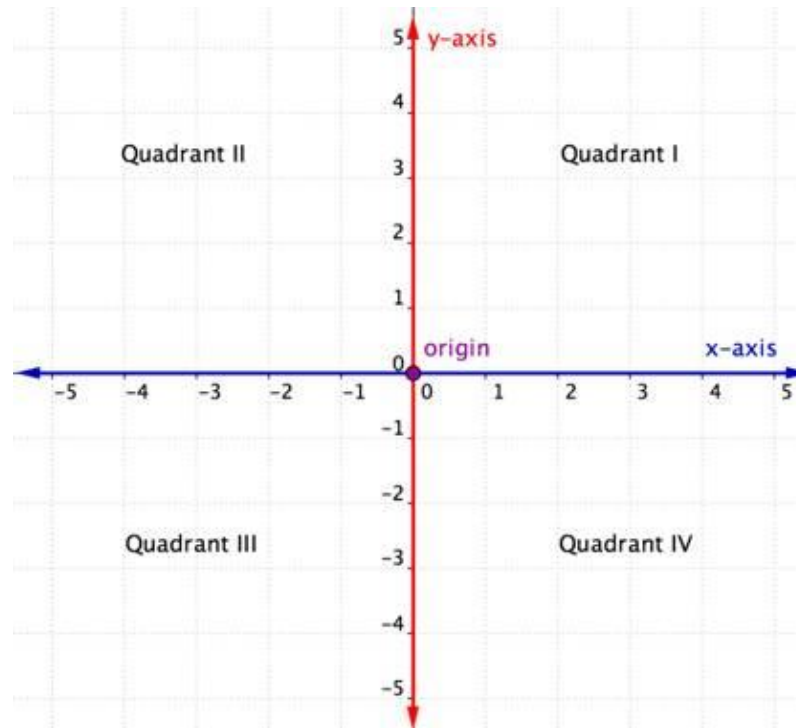
Tilt the floor using keyboard keys

- The W key rotates the floor by 5 units forward and the S key rotates the floor by 5 units backward (Use the Y axis).
- Add two 'Add Actor World Rotation' to the floor actor.
- Connect the input node Pressed pin to the 'Add Actor World Rotation' input pin.

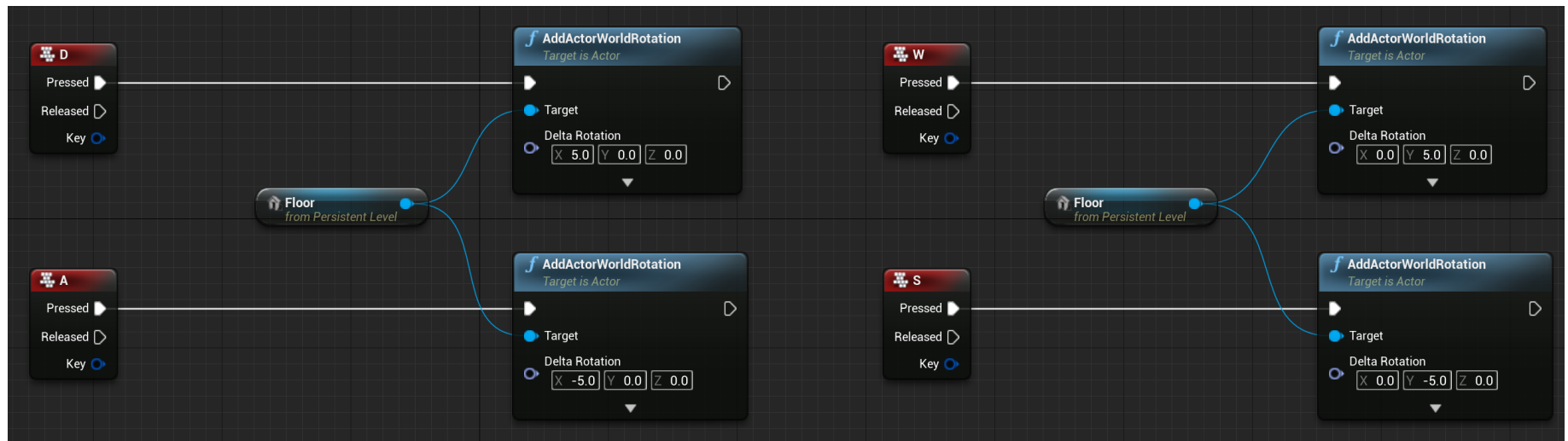
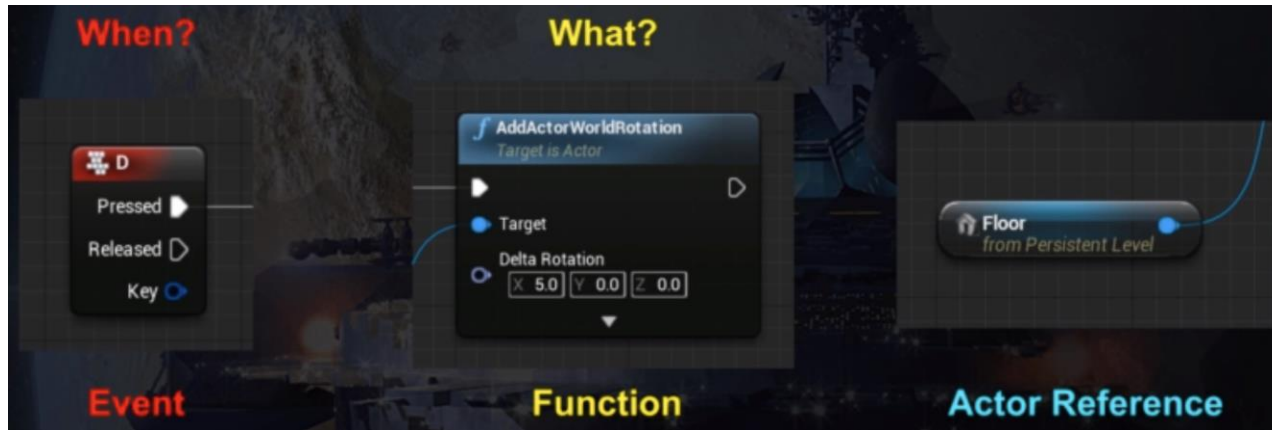


Tilt the floor using keyboard keys

In transform related functions, positive values are used for right and forward directions, while negative values are used for left and backward directions.

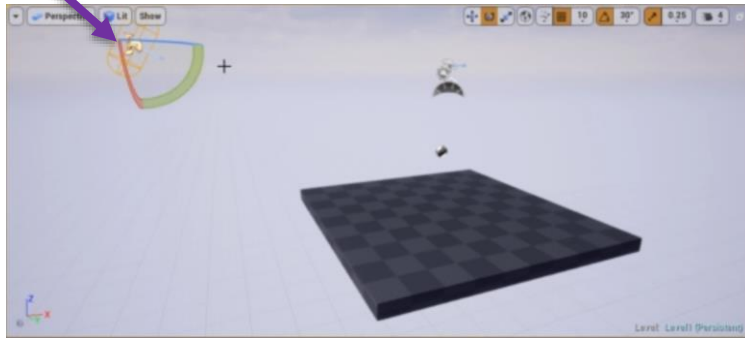


Tilt the floor using keyboard keys

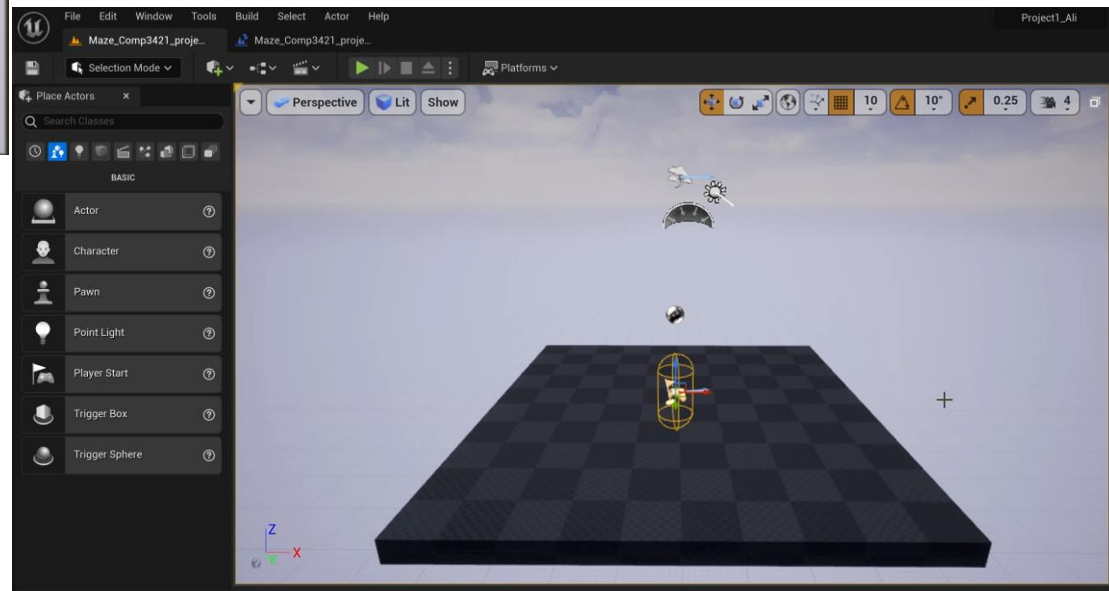


Change the player start position

- In order to have a better view of the maze game change the start position of the player start.



If you want to restart your view, select your desired object and press 1 key.



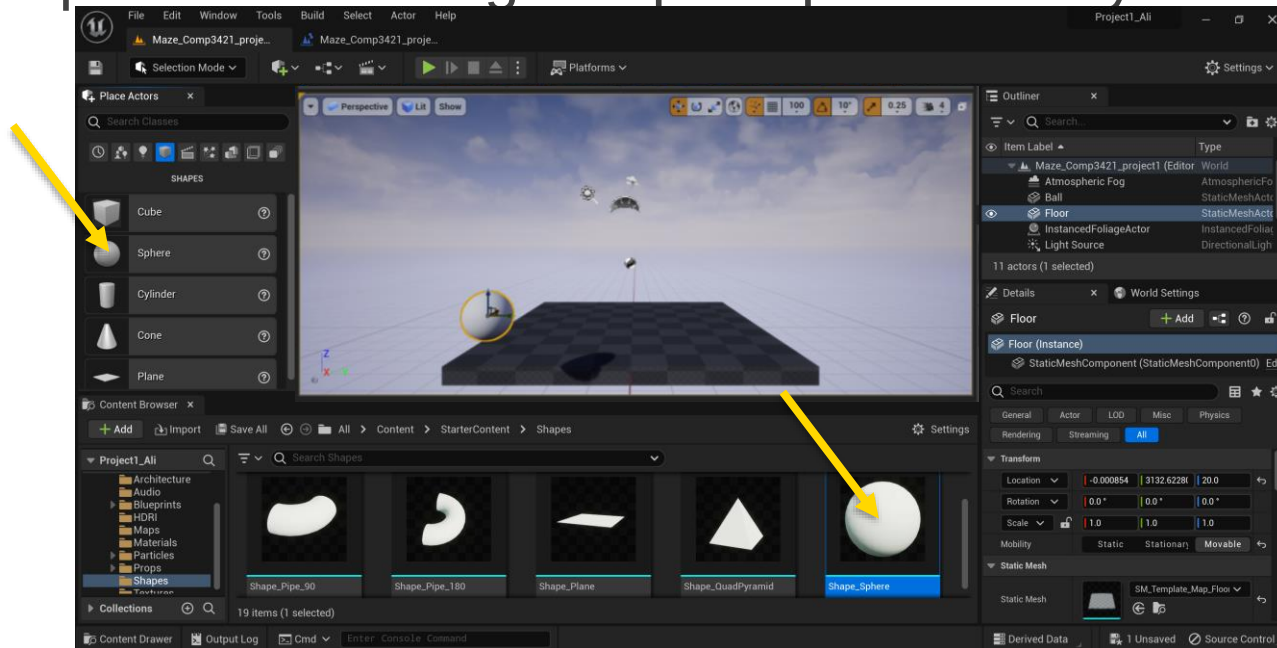
Lecture activity 1

- Create a blank project in Unreal Engine and add a new default level.
- Create a blueprints code so that we can tilt the floor to four different directions.
- By pressing W key, we want to pitch the floor actor forward.
- By pressing S key, we want to pitch the floor actor backward.
- By pressing A key, we want to pitch the floor actor to the left.
- By pressing D key, we want to pitch the floor actor to the right.



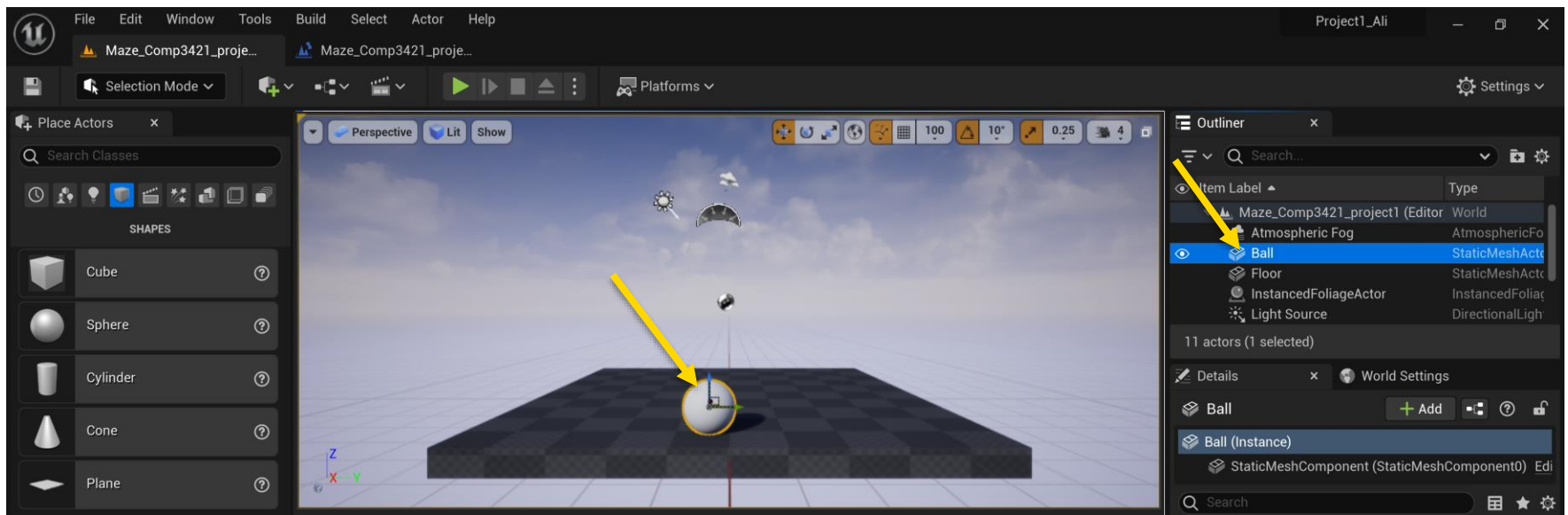
Add content to the level (Add the maze ball)

- Go to the **Place Actors** panel > Basics > Add a Sphere by dragging it into the Viewport.
- Or go to the assets content browser > Starter content folder > Shape folder > Drag Shape Sphere into your level.



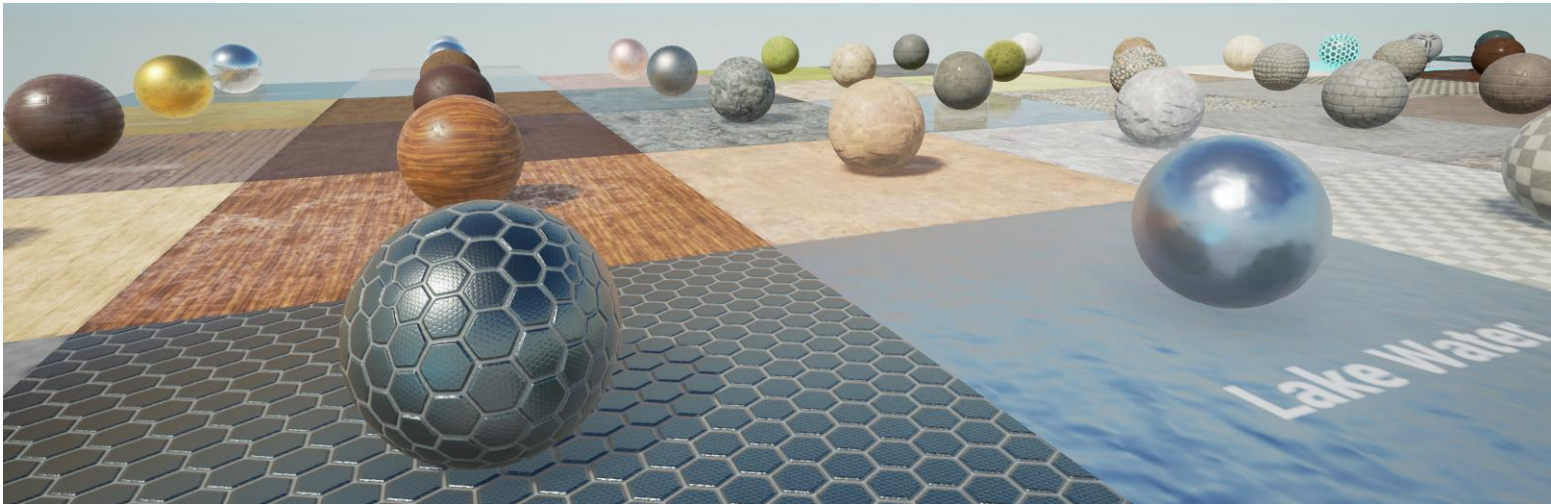
Add content to the level (Add the maze ball)

Rename the sphere name to ball from the World Outliner panel.



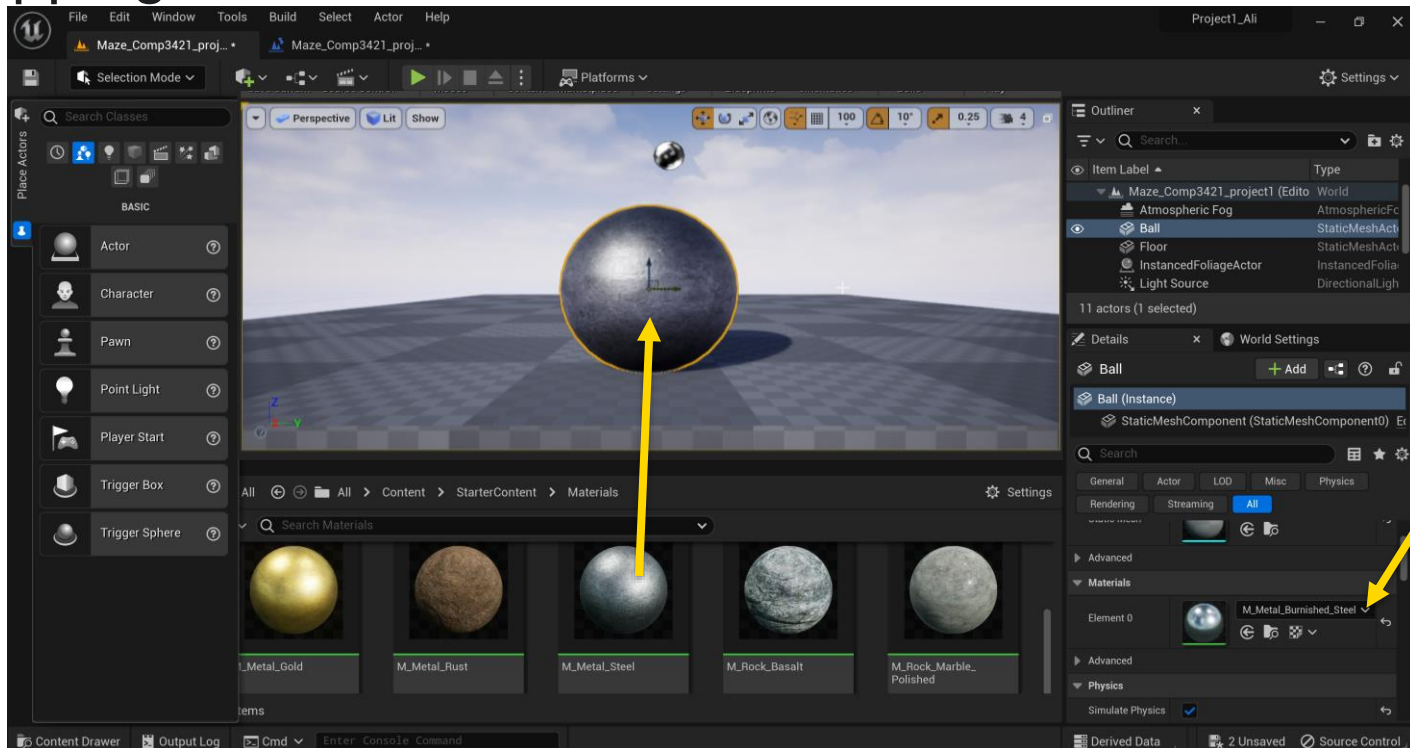
Add material to the objects

A Material is an asset that can be applied to an object to control the visual look of the scene.



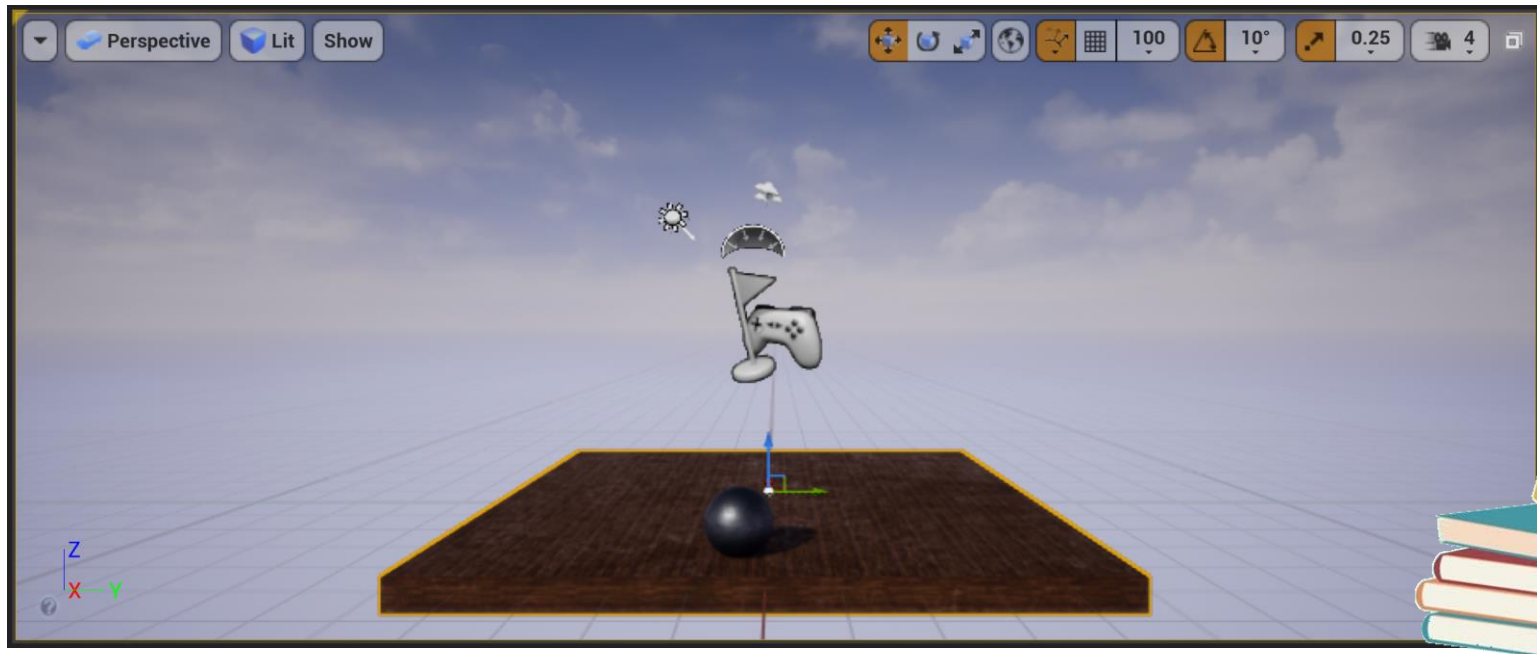
Add material to the objects

- Change the object's material by selecting a material from the dropdown menu of the material in the detail panel.
- Or by dragging the material from the content browser and dropping it on the actor.



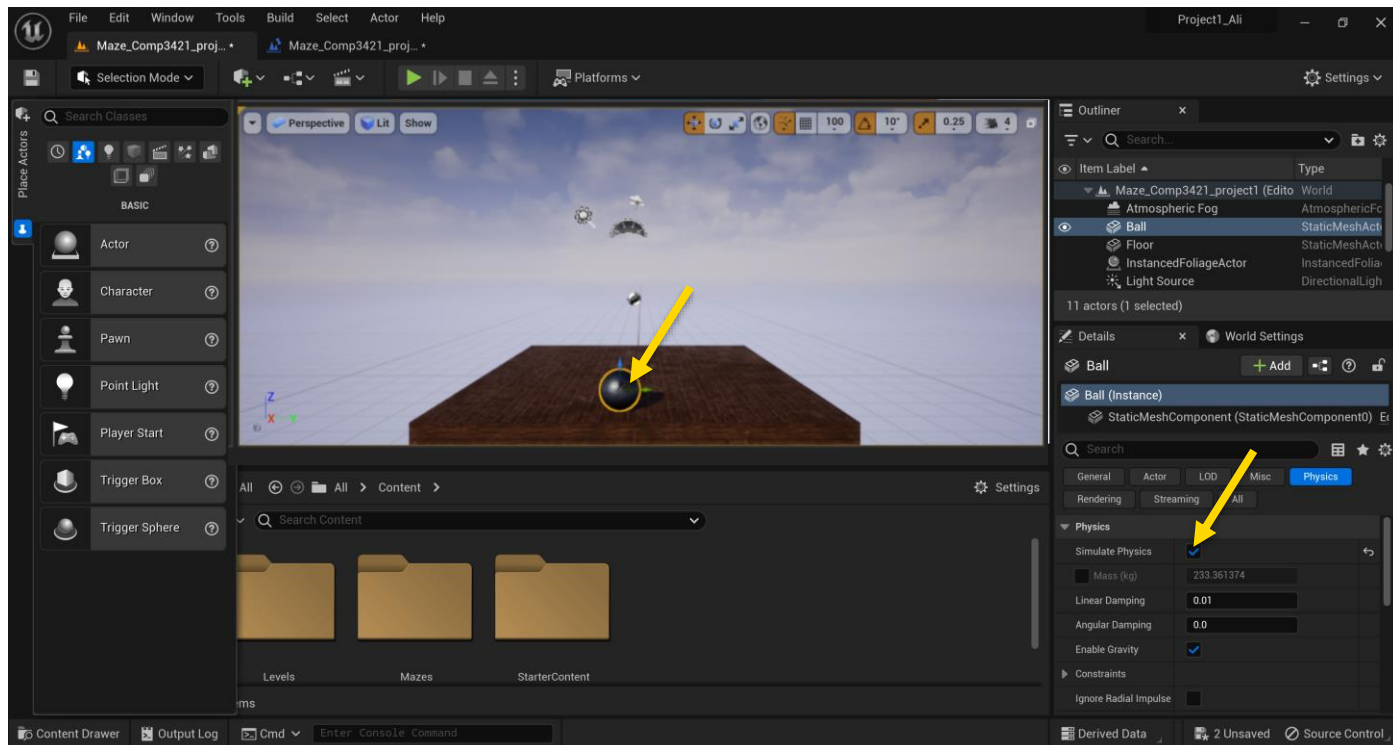
Lecture activity 2

- Add a metal like material to the ball.
- Add a wooden like material to the floor.

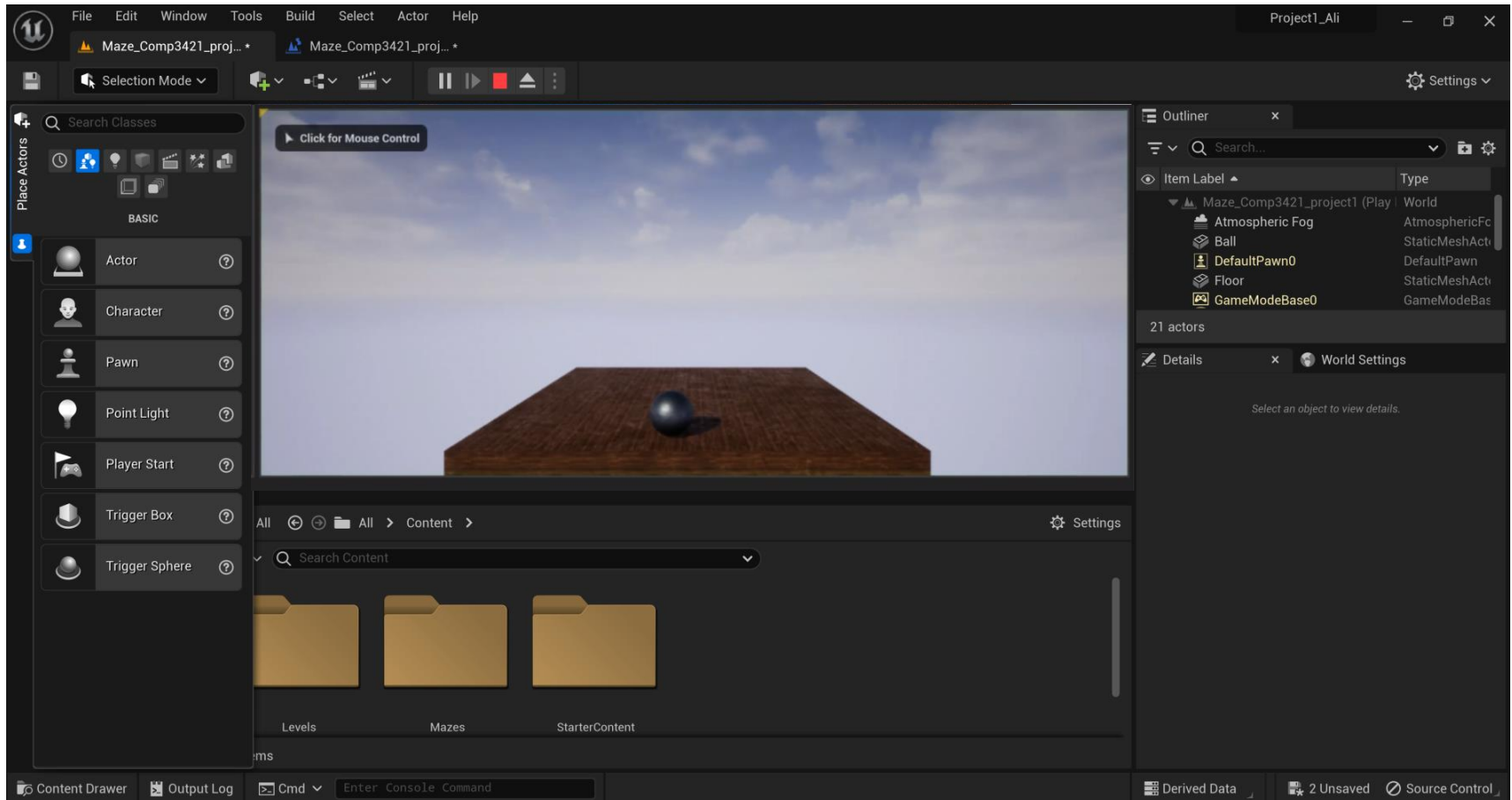


Applying physics to the ball

- **Simulate Physics** feature adds different physical responses to an object such as Gravity, Force, Impulse, and Physical Constraints.
- Details panel > Physics section > Simulate Physics.

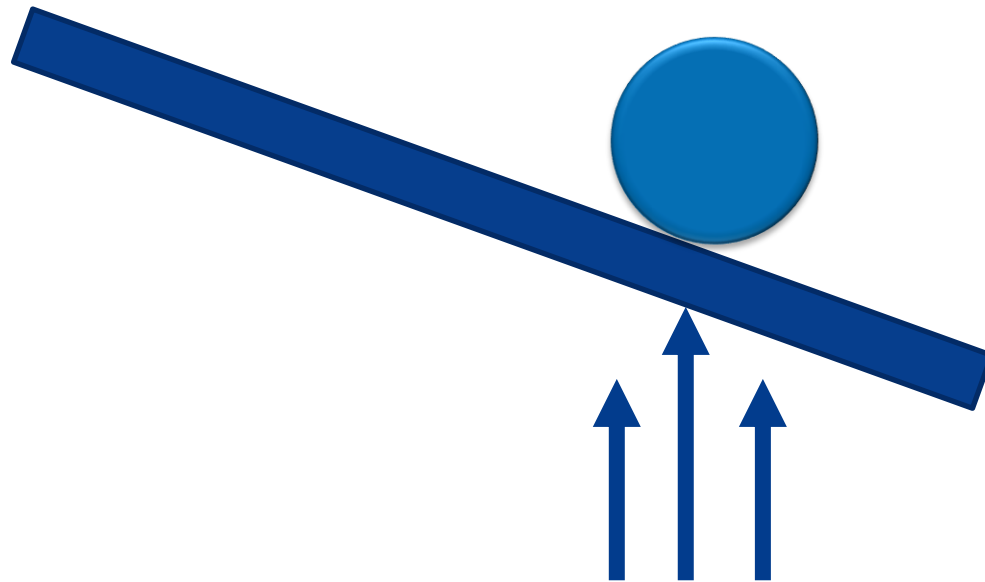


Applying physics to the ball



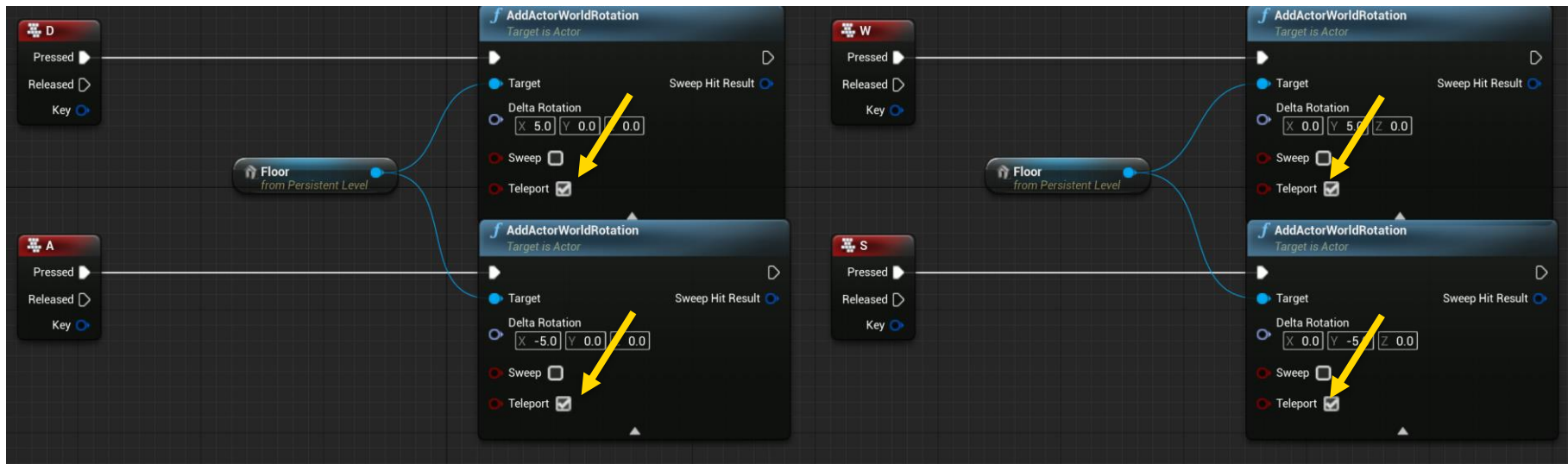
Applying physics to the ball

- The ball is thrown in the air instead of moving to the left, because the physics velocity of the floor is changing and applies an upward force to the ball.
- We should tell Unreal to ignore force being applied to the ball.
- We should teleport physics state of the floor.

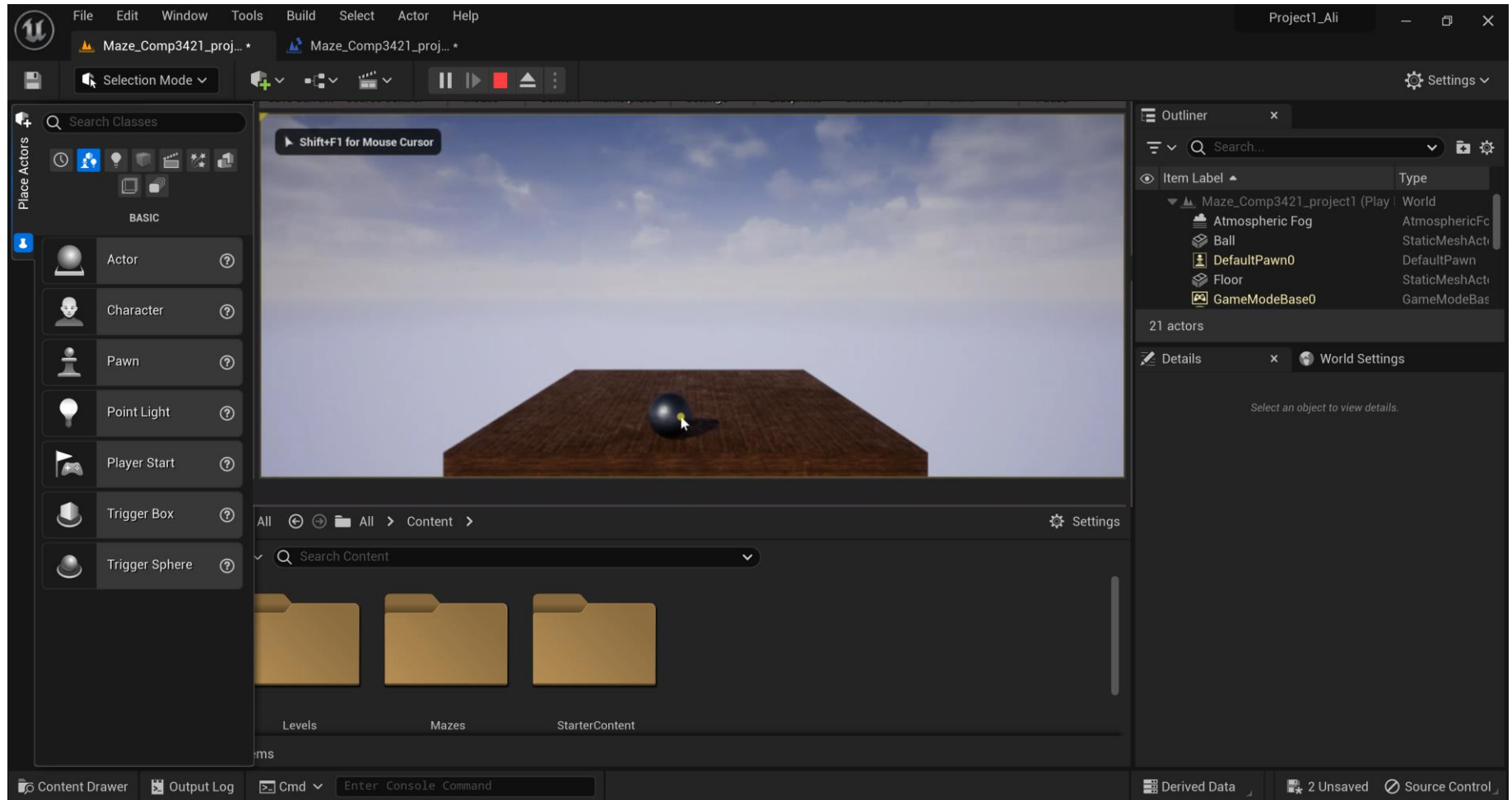


Applying physics to the ball

Go to the level blueprint > Expand 'Actor World Rotation' function > Make 'Teleport' active.

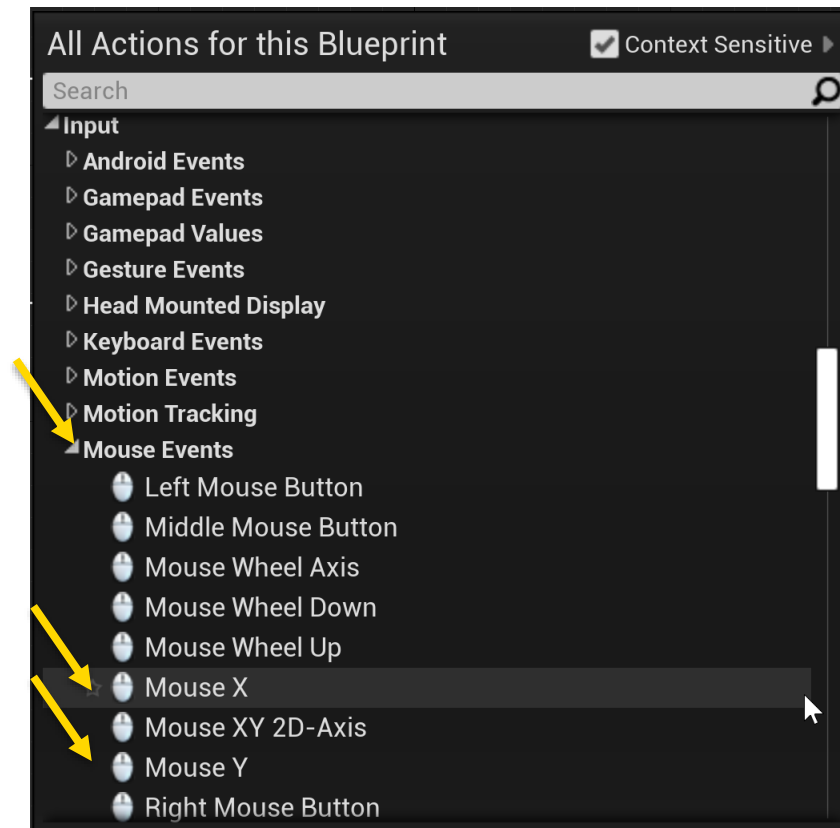


Applying physics to the ball



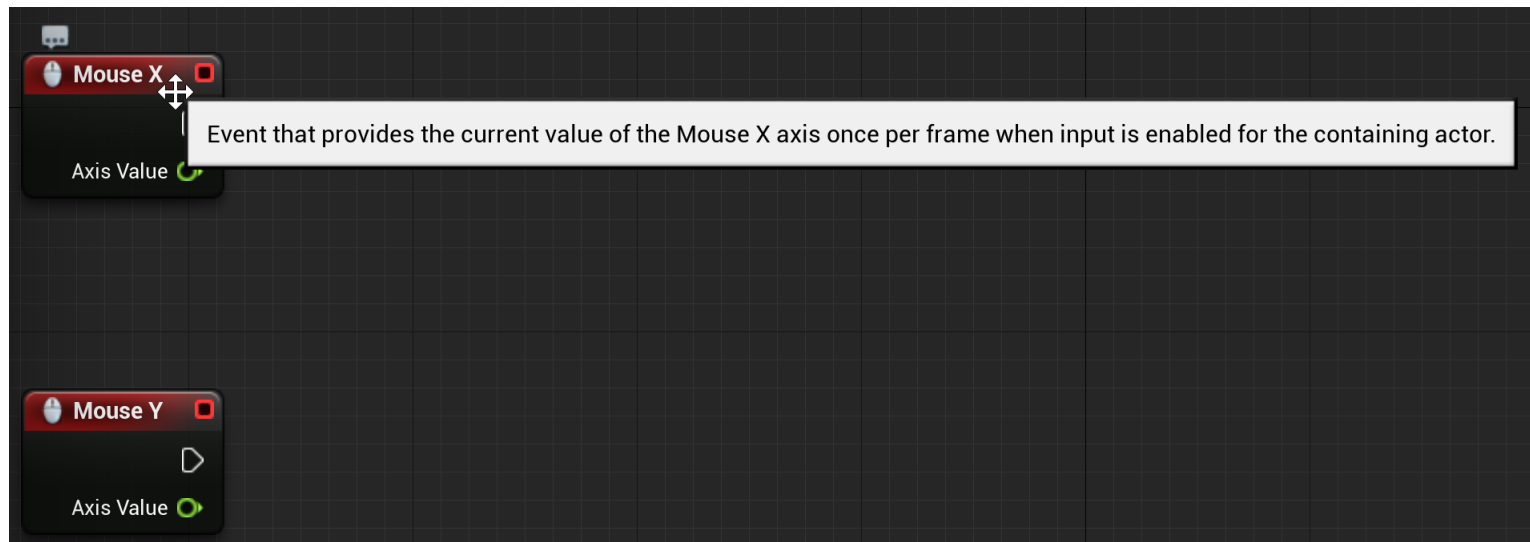
Tilt the floor with mouse input

- Go to the Input category > Mouse Events > Mouse X & Mouse Y



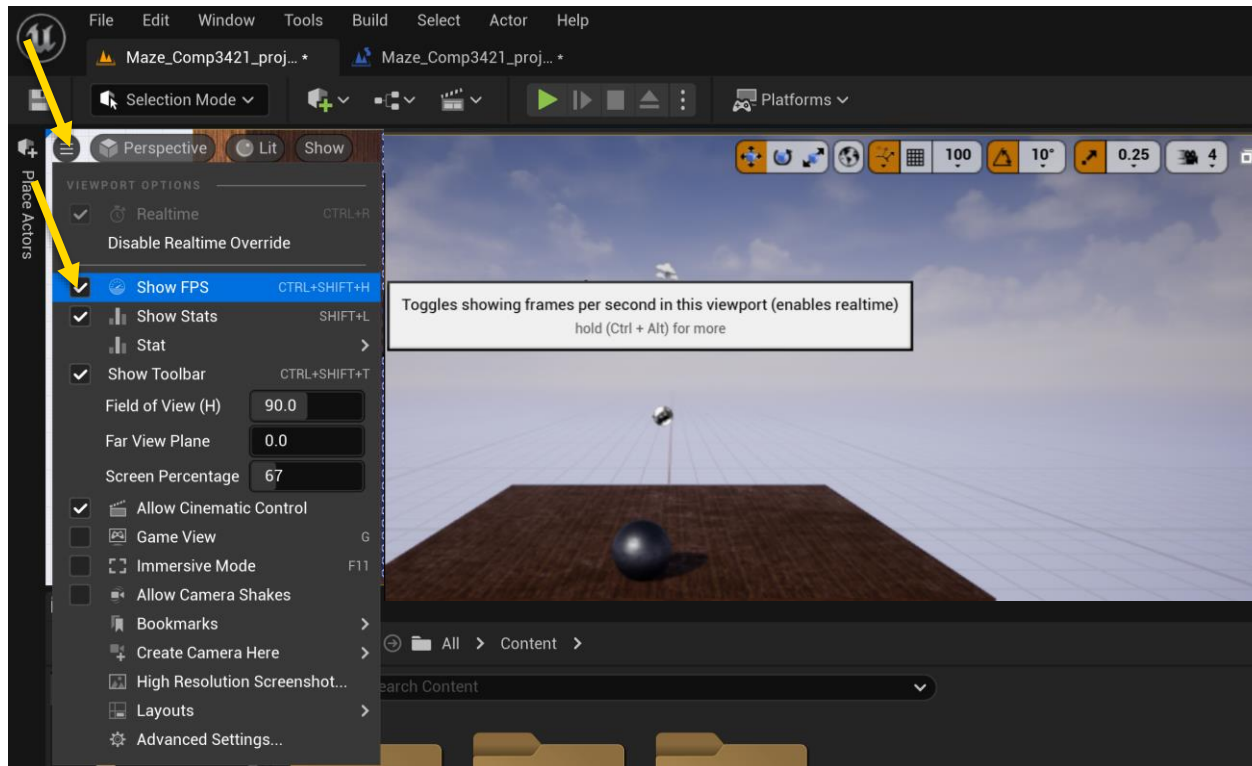
Tilt the floor with mouse input

- Mouse X and Mouse Y events work, when they receive inputs
- They are updated every frame and return the X and Y axes values of mouse.



Tilt the floor with mouse input

- In order to view frames per second check 'Show FPS'.

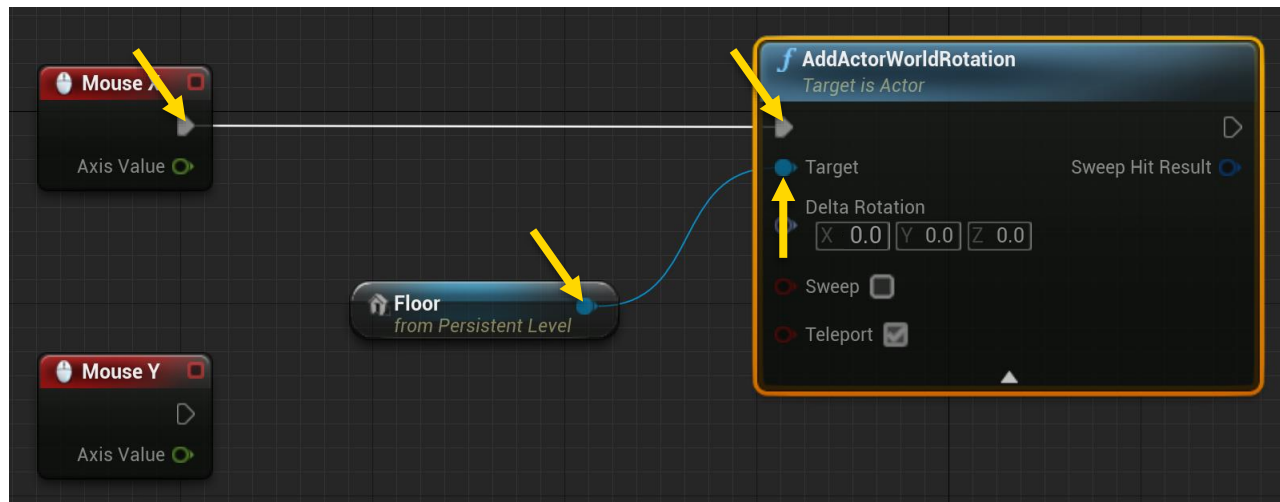


Tilt the floor with mouse input

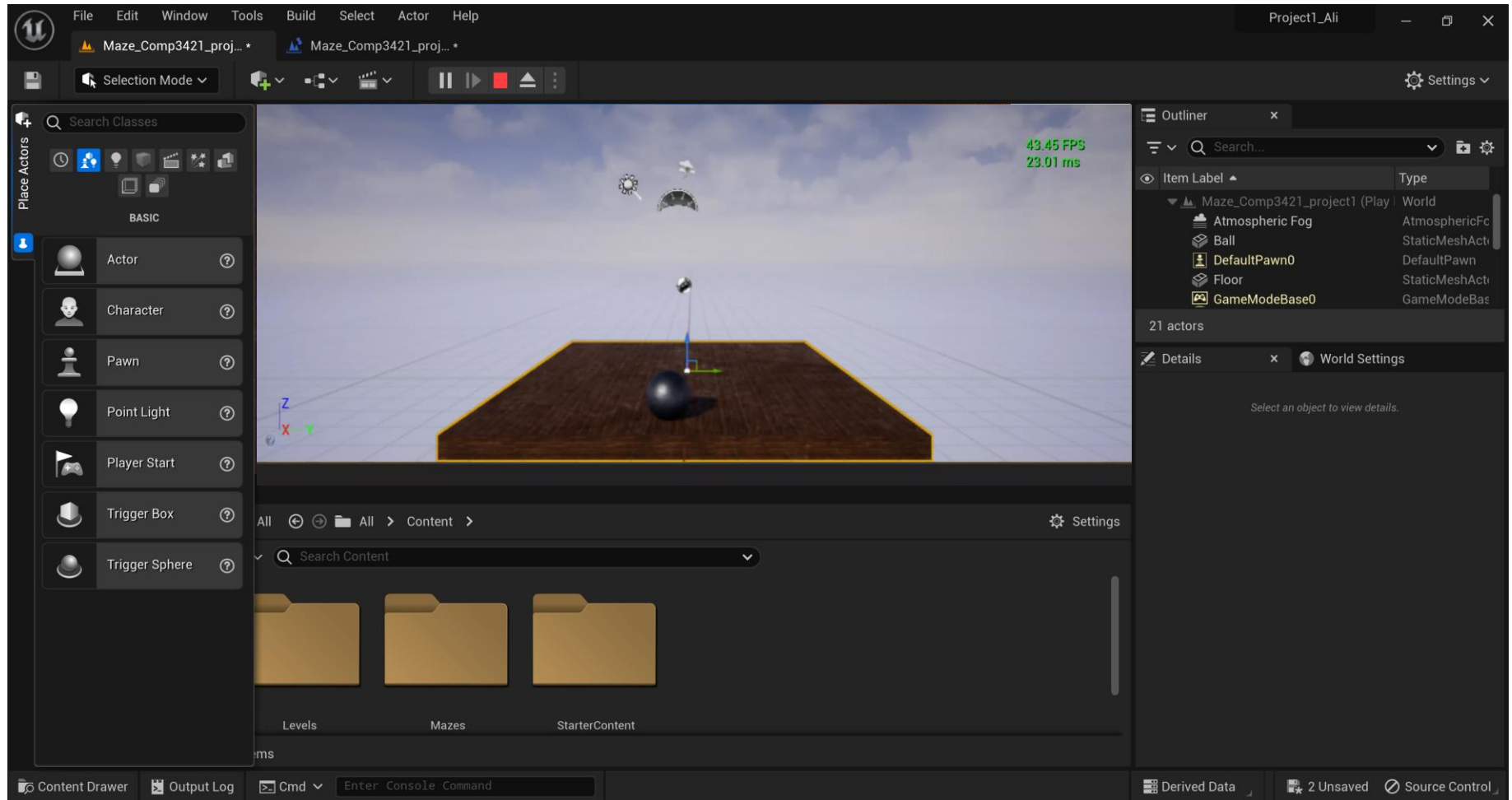


Tilt the floor with mouse input

- Add the Floor actor and 'Add Actor World Rotation' function.
- Connect the execute pin of the mouse to 'Add Actor World Rotation' function.
- Connect the Floor actor to the target pin of 'Add Actor World Rotation' function.

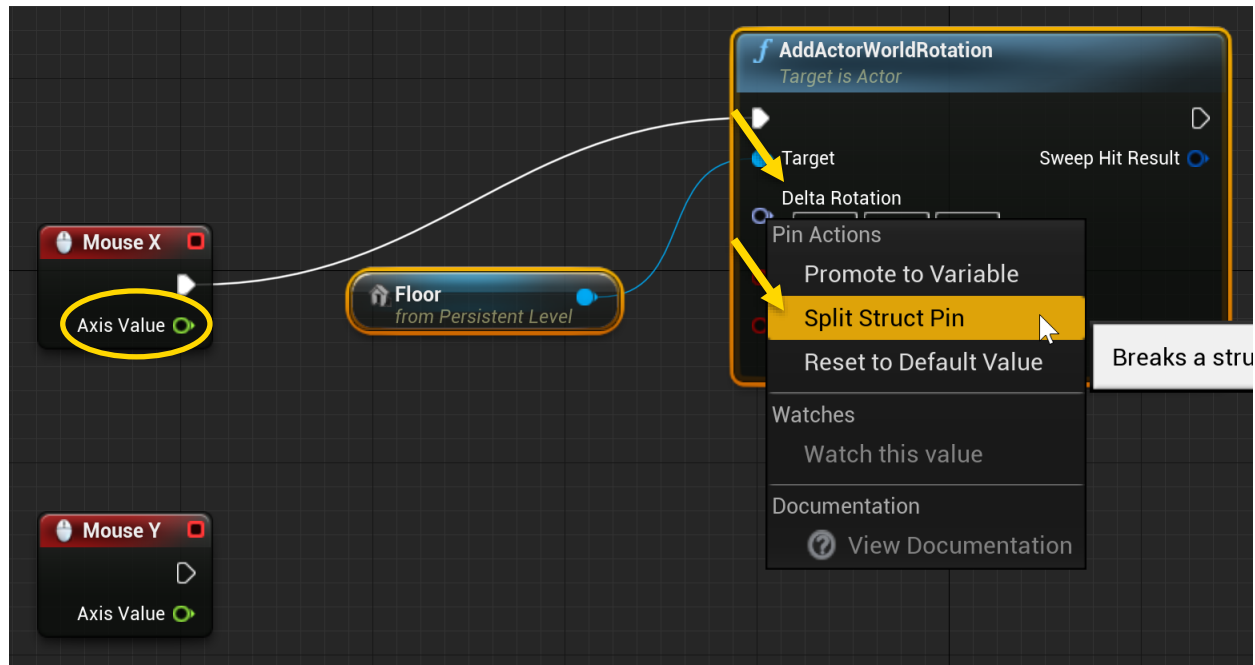


Tilt the floor with mouse input



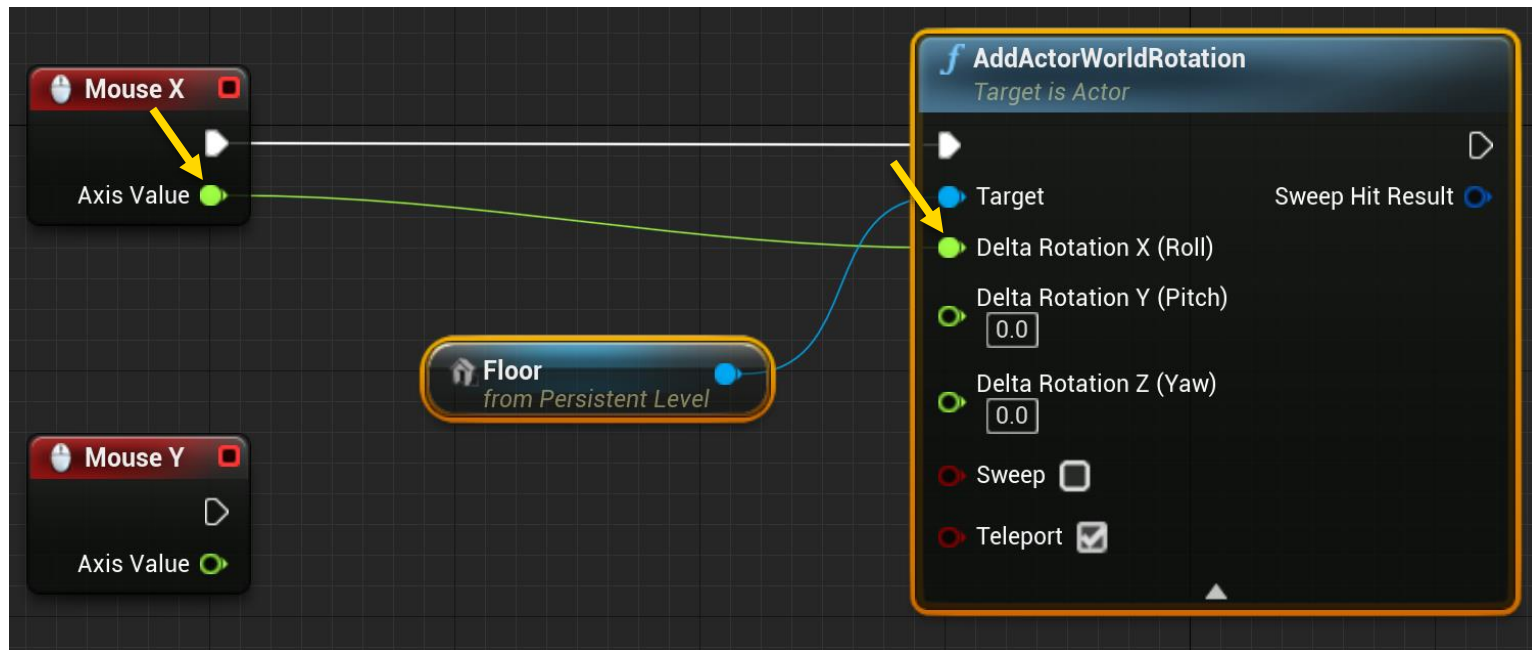
Tilt the floor with mouse input

- Connect the **Axis Value** pin of Mouse X node to '**Add Action World Rotation**' function.
- Right click on '**Delta Rotation**' pin > Click '**Split Struct Pin**'.

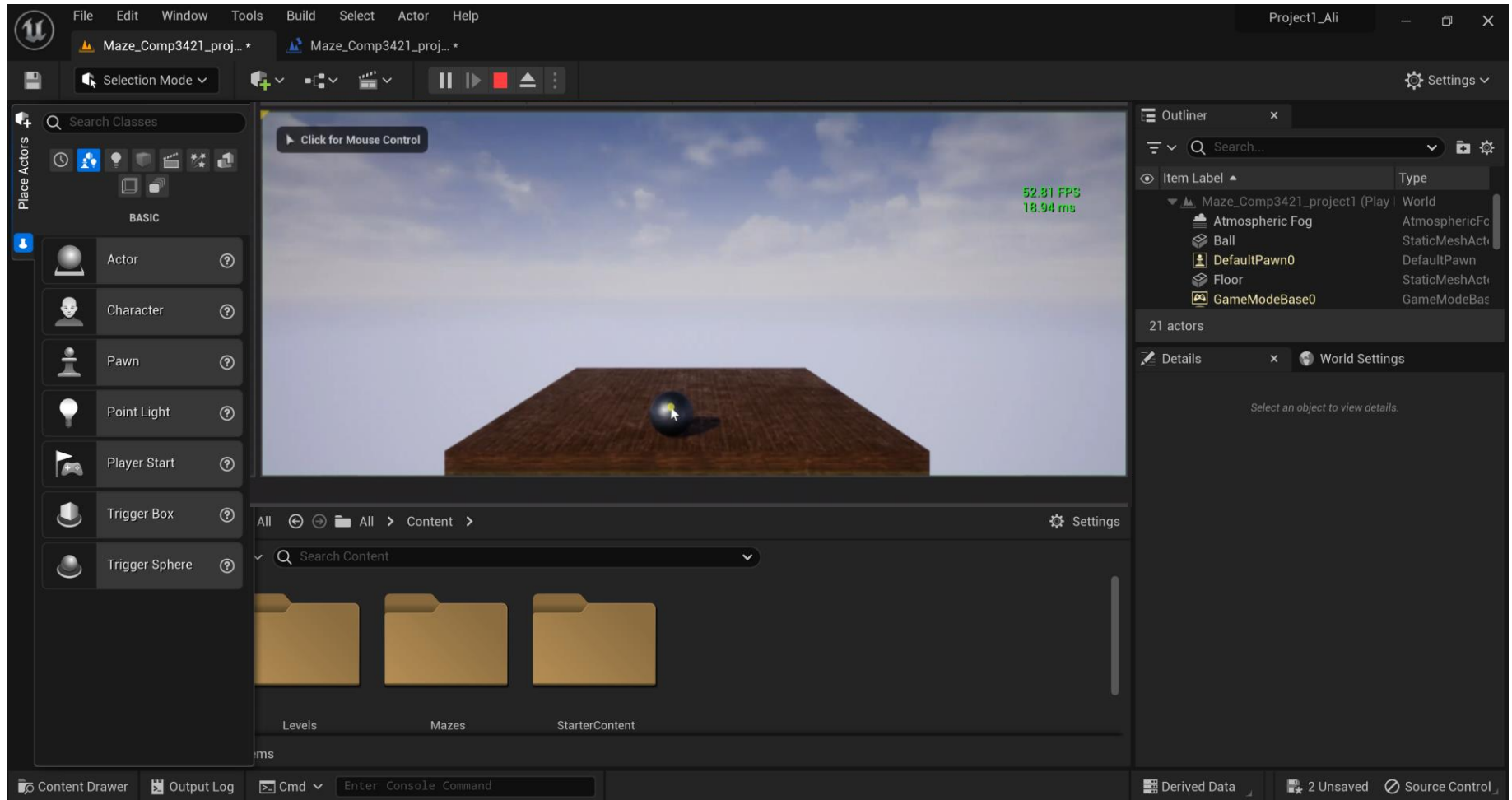


Tilt the floor with mouse input

- Plug the **Axis Value** pin of **Mouse X node** straight into the **Delta Rotation X (Roll)**.



Tilt the floor with mouse input



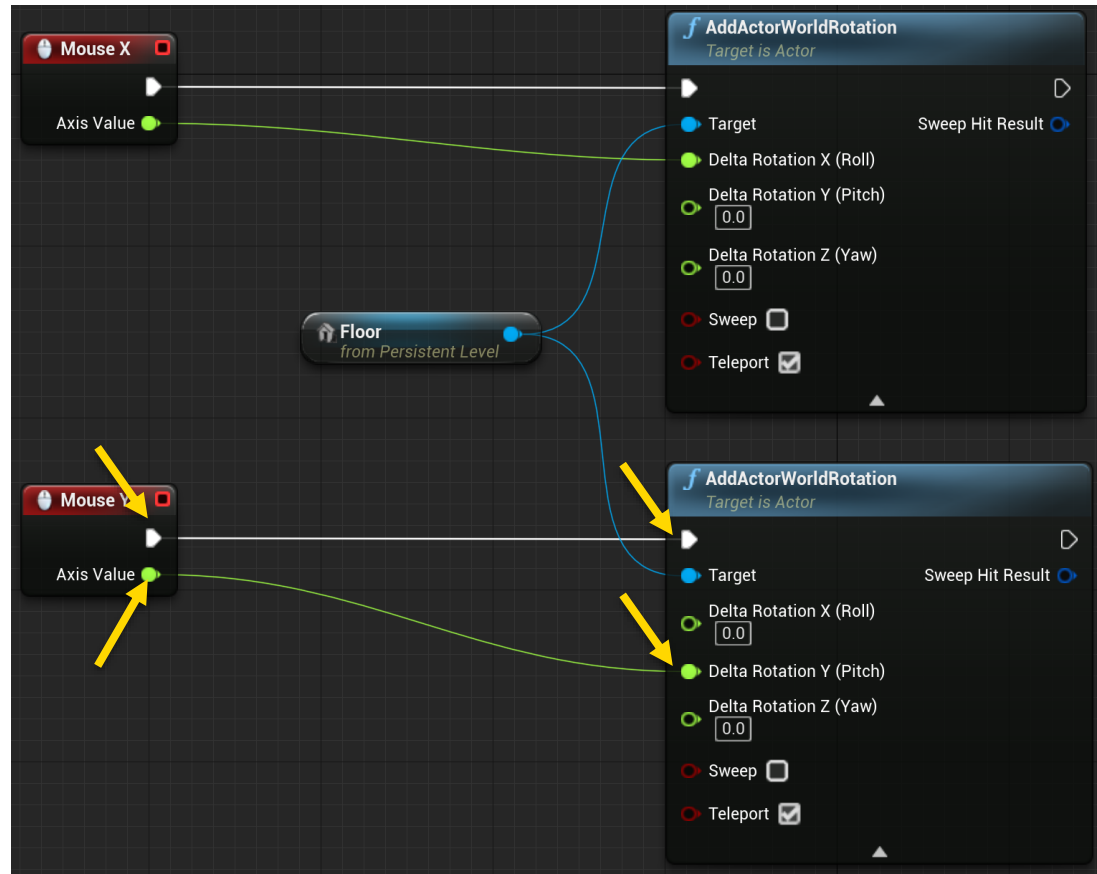
Lecture activity 3

Add a blueprint code to control the Roll of the floor actor by mouse movement to tilt it left and right.



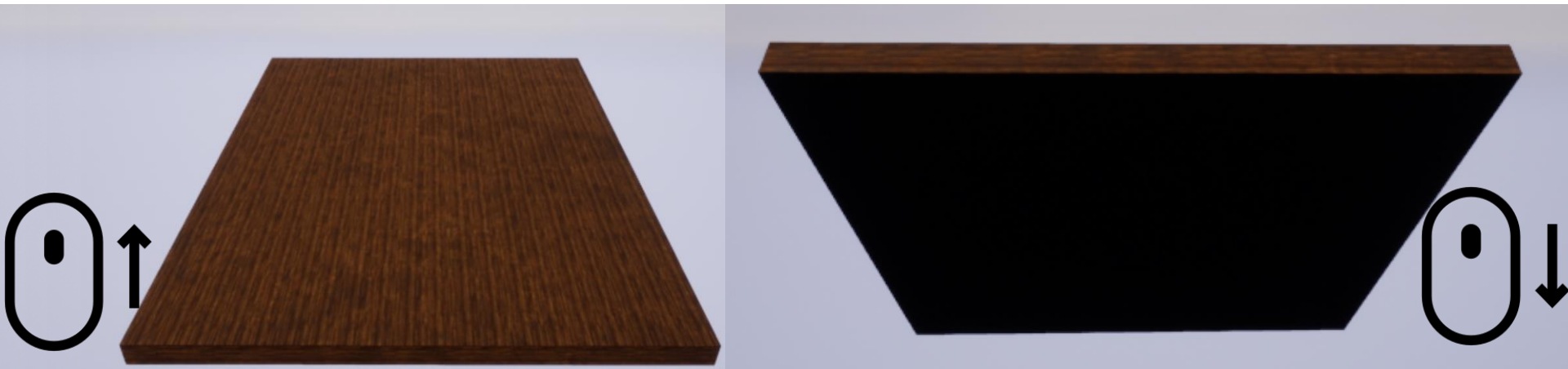
Tilt the floor with mouse input

- Add another 'Add Actor World Rotation' function.
- Connect the Axis Value pin of Mouse Y node to 'Add Action World Rotation' function.
- Right click on 'Delta Rotation' > Click 'Split Struct Pin'.
- Plug the axis value of Mouse Y straight into the Delta Rotation Y (Pitch).



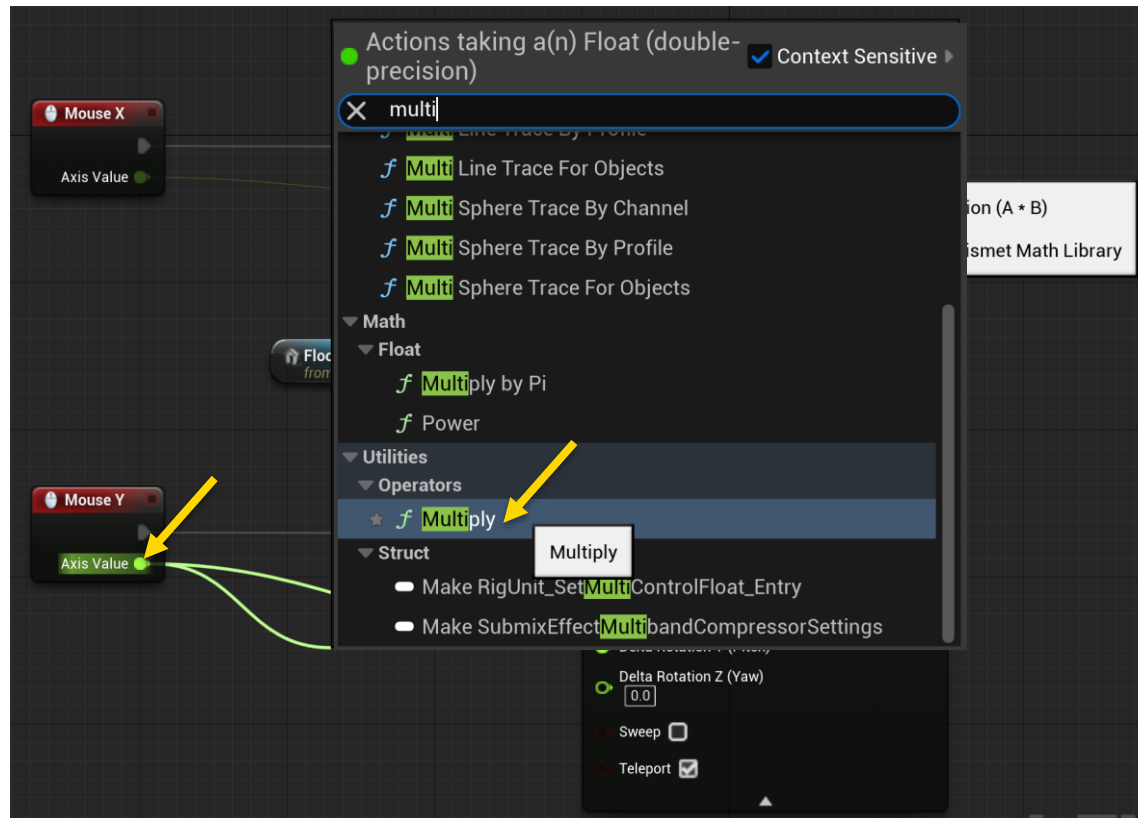
Tilt the floor with mouse input

There is an issue: When we move mouse up the floor tilt down and when we move mouse down, the floor tilt up.



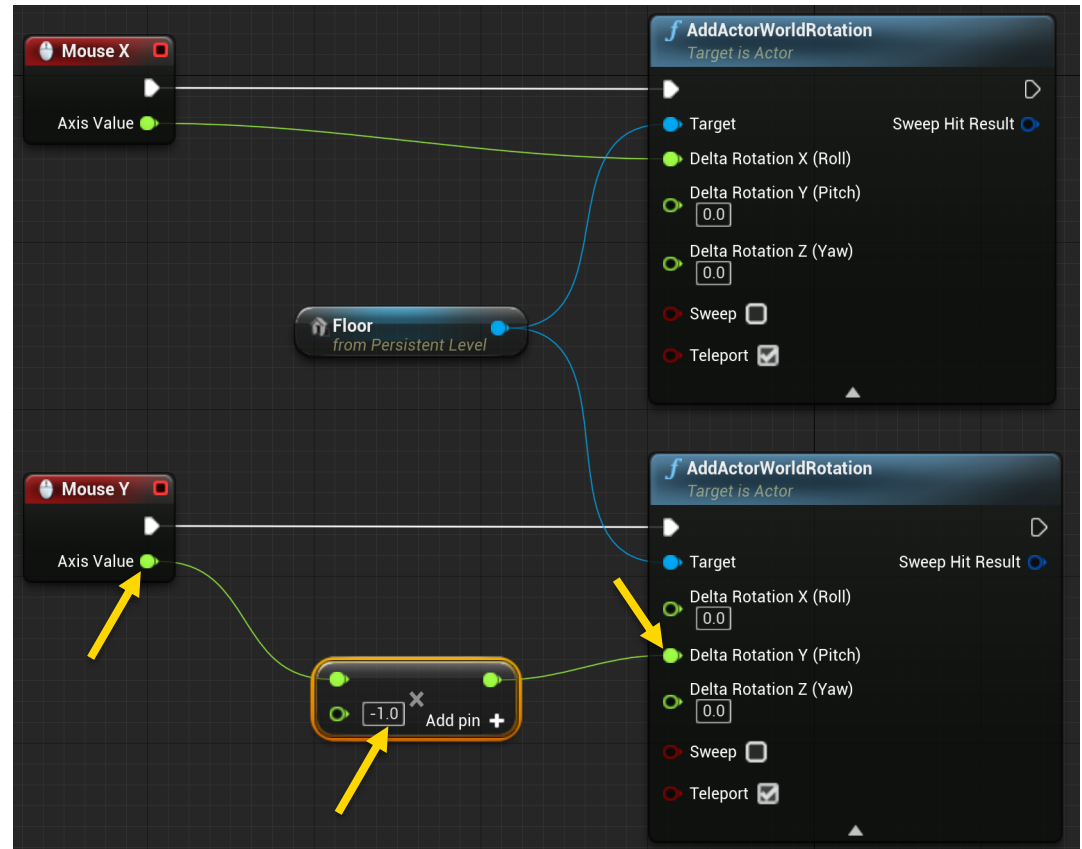
Tilt the floor with mouse input

- To solve the reverse movement issue, the mouse value should be multiplied by -1.
- Drag out from **Axis Value** pin of **Mouse Y** node > Add **Multiply** function.

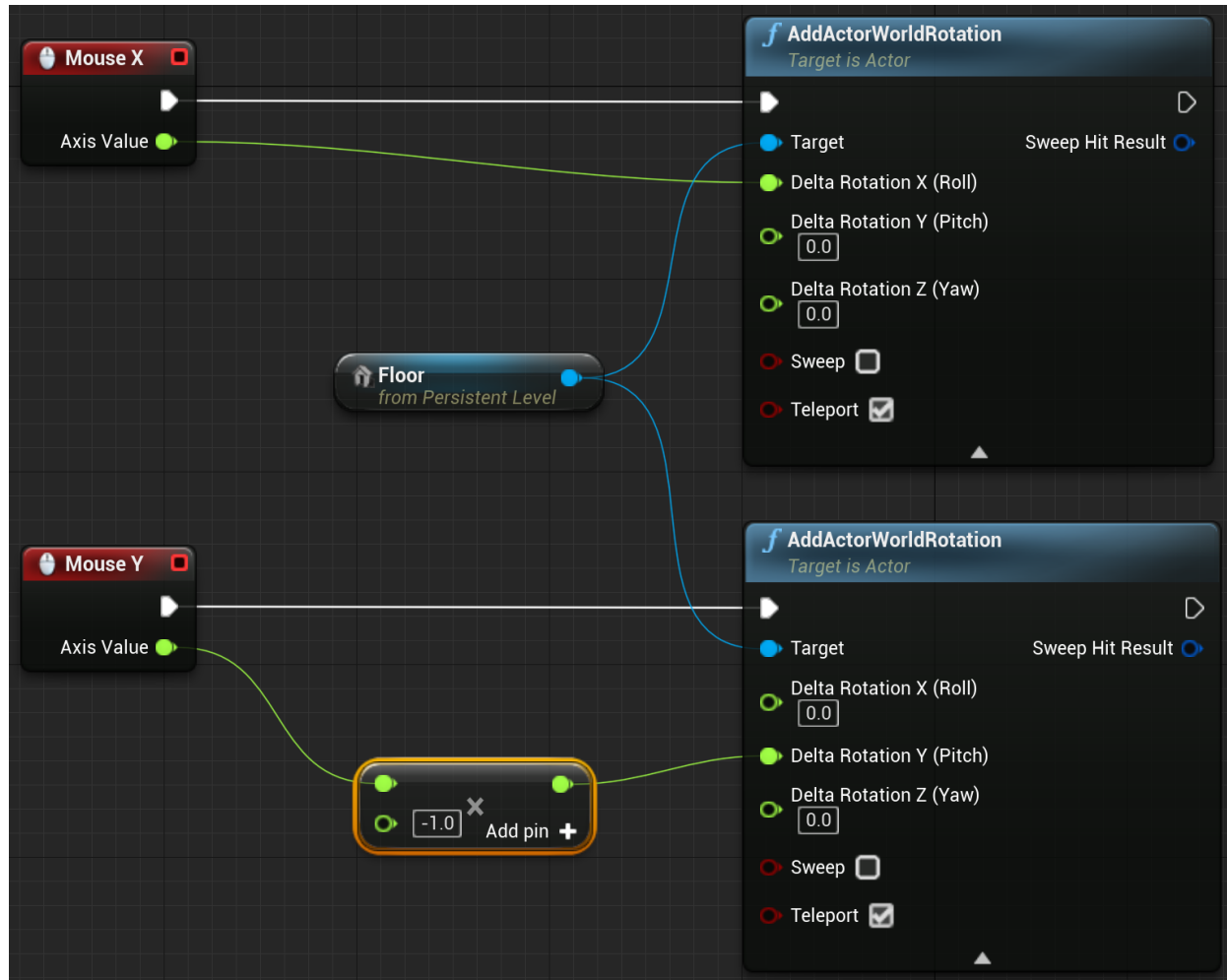


Tilt the floor with mouse input

- Connect **Multiply** function to **Delta Rotation Y**.
- Type -1 in the multiply function in order to invert mouse movement direction.



Tilt the floor with mouse input (Summary)



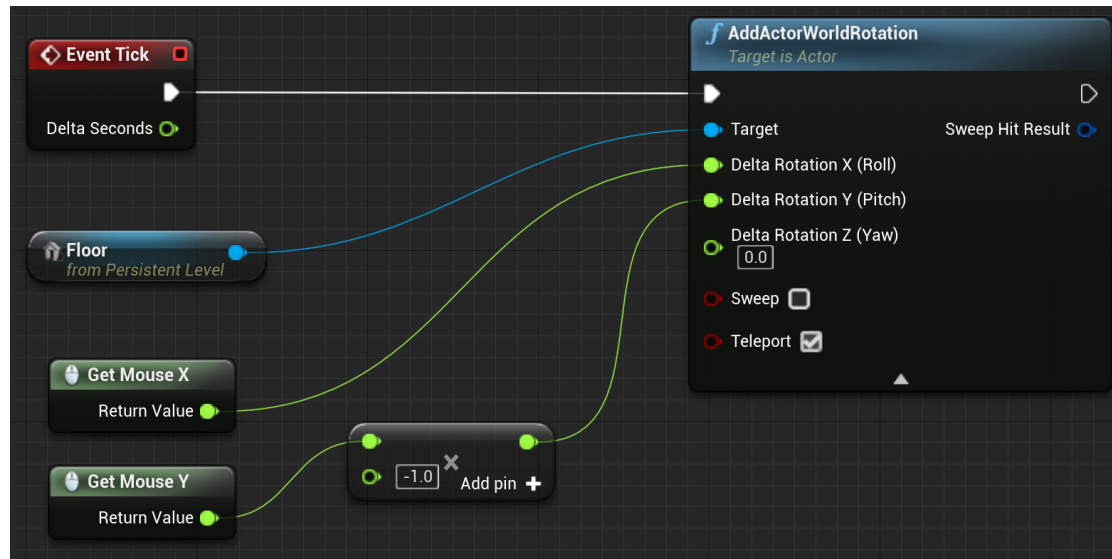
Lecture activity 4

- When we move mouse up the floor should tilt up and when we move mouse down, the floor should tilt down.
- Solve the issue of mouse movement by inverting the mouse movement direction.



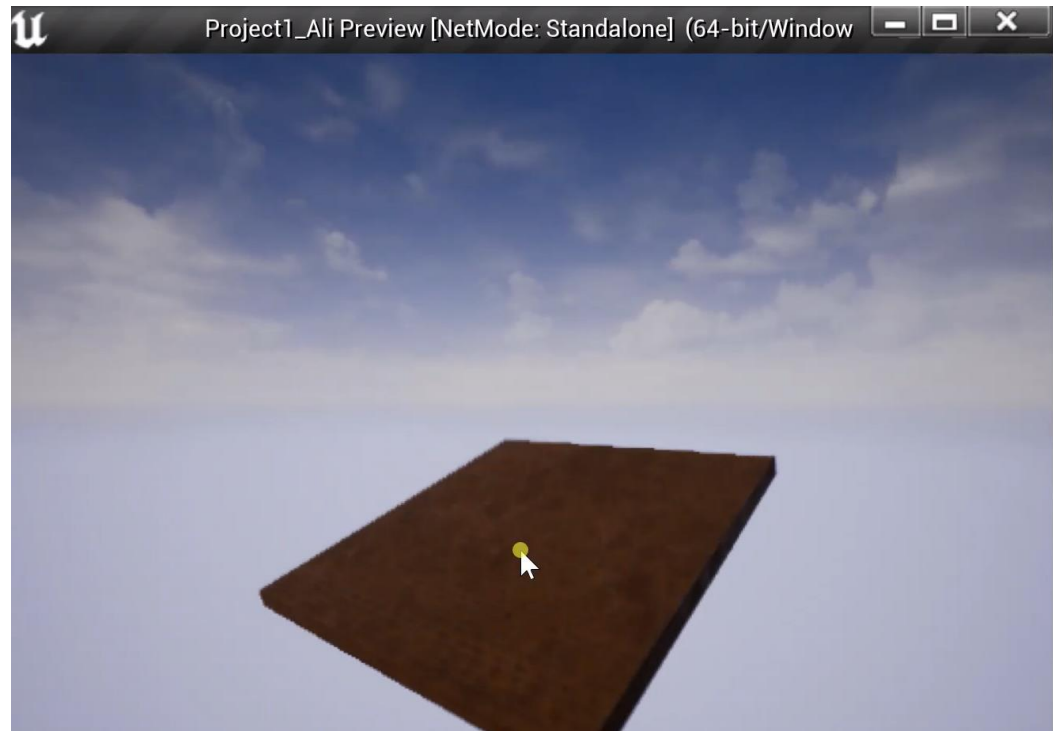
Another way to tilt the floor with mouse input

- Use **Event Tick** that is called every frame. It is same as the way that mouse X and mouse Y work.
- Get mouse position using '**Get Mouse X**' and '**Get Mouse Y**' nodes.
- Connect **Event Tick** to the input pin of the '**Add Actor World Rotation**' function > Floor to the target pin of the '**Add Actor World Rotation**' > **Get Mouse X** node to the **Delta Rotation X** pin > **Get Mouse Y** node to the **Delta Rotation Y** pin.



Tilt the floor with mouse input (Offset issue)

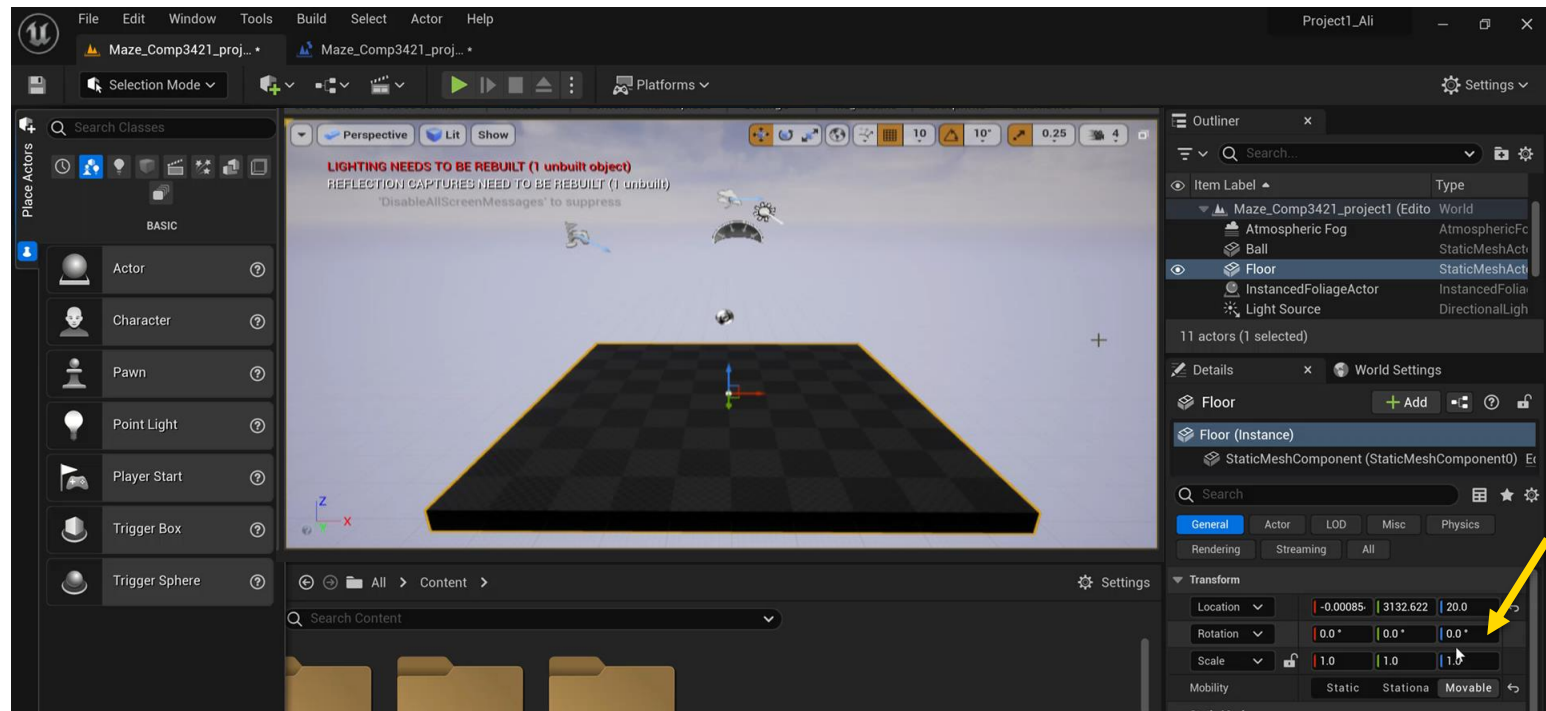
When we add pitch to the floor and when we add a roll and then if we pitch the floor back and then roll it back, we can see that the floor starts rotating.



Tilt the floor with mouse input (Offset issue)

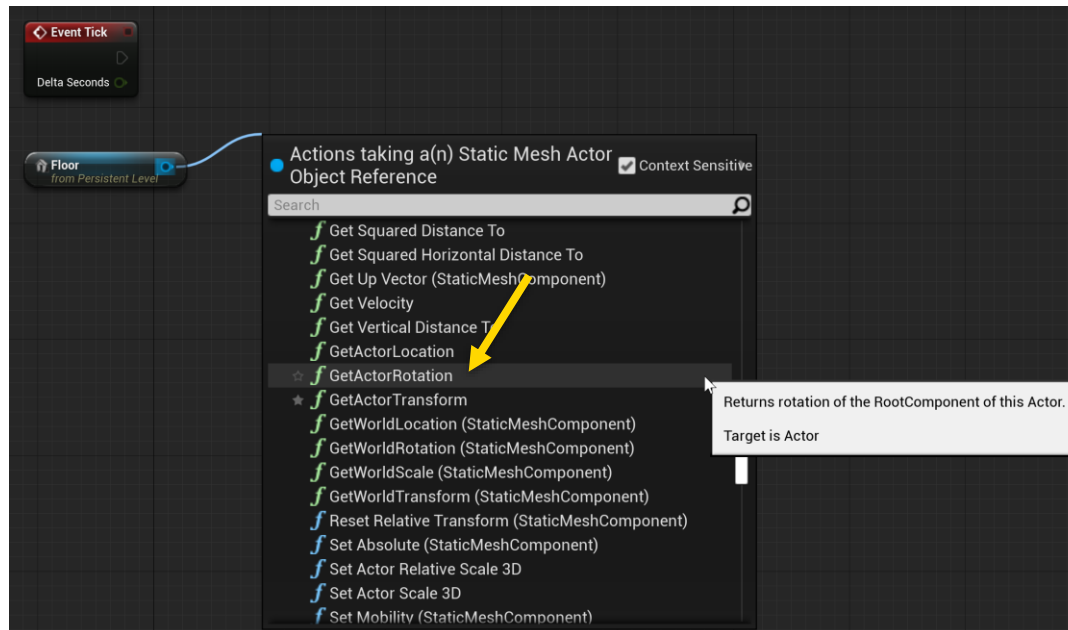
Solution:

- Get rotation of the floor.
- Set new rotation of the floor actor with mouse input.
- Filter out Yaw rotation (**The Z axis should be kept at 0**).



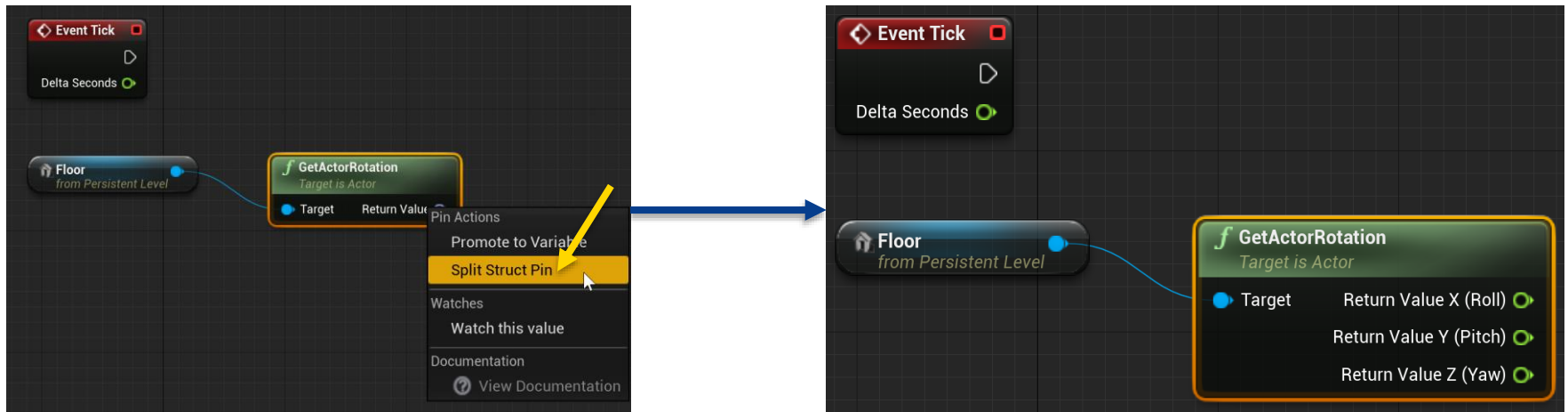
Tilt the floor with mouse input (Offset issue)

- Instead of using 'Add Actor World Rotation' function, use another function called 'Get Actor Rotation'.
- **Add Actor World Rotation** function adds a delta (X, Y ,Z) to the rotation of the actor in the world space.
- **Get Actor Rotation** function returns the current actors' rotation.
- **Set Actor Rotation** function sets the Actor's rotation instantly to the specified rotation.



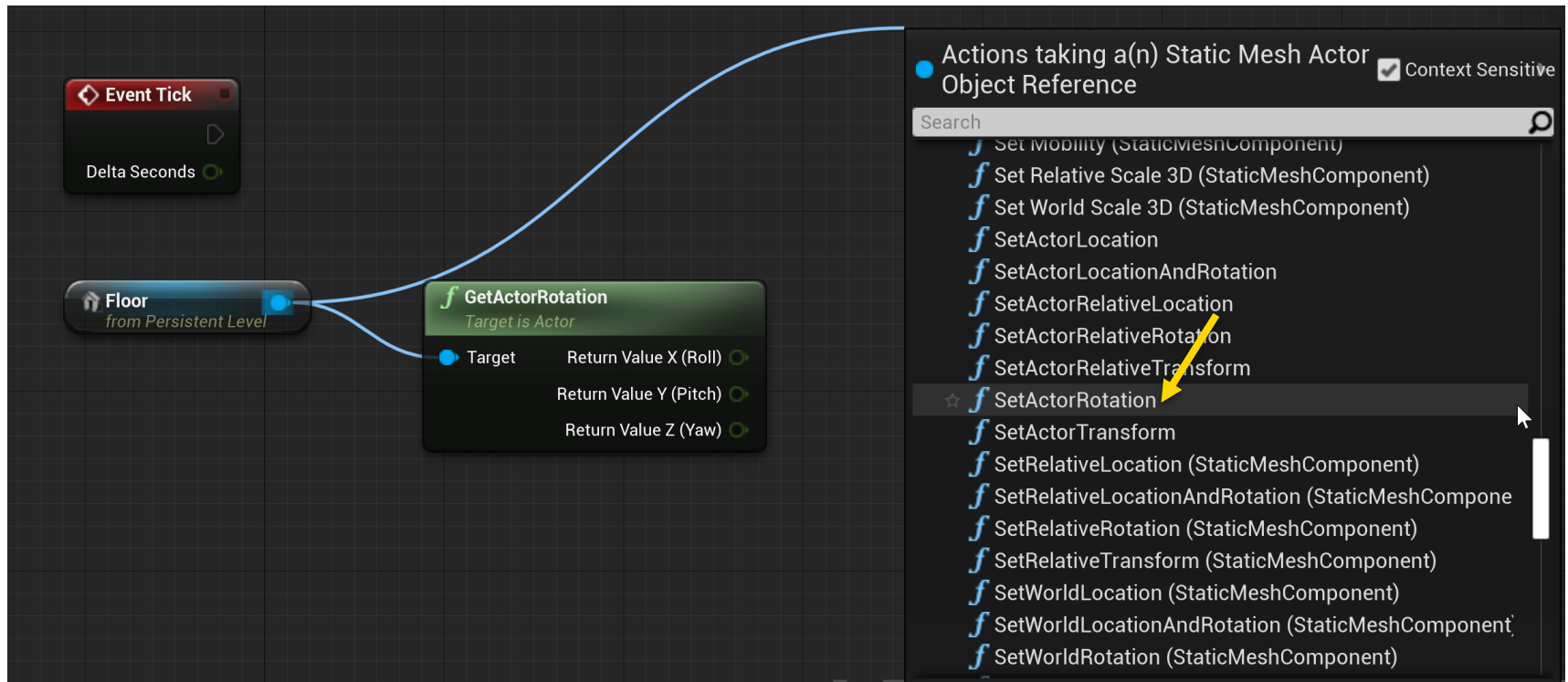
Tilt the floor with mouse input (Offset issue)

Split the **Return Value** pin of 'Get Actor Rotation' function to gain access to the Roll, pitch and Yaw separately.



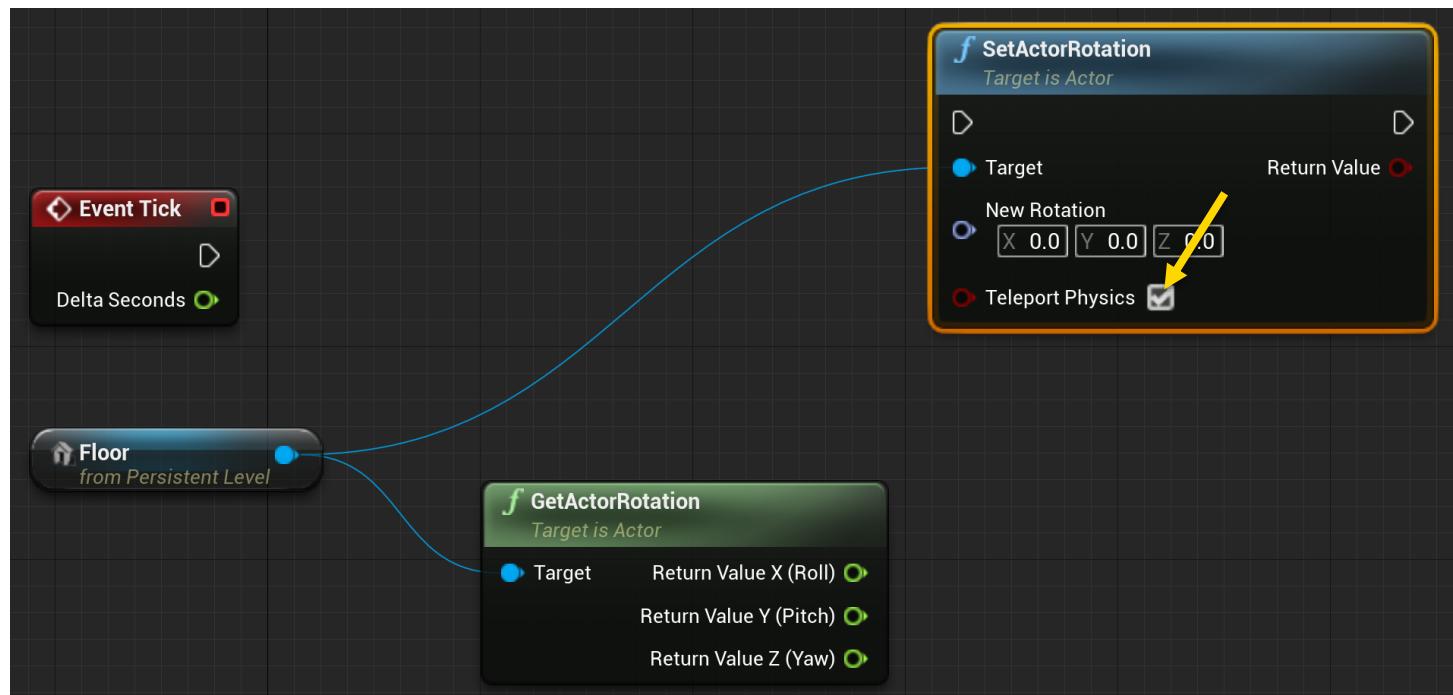
Tilt the floor with mouse input (Offset issue)

- Drag out from the **Floor** node > Utility > Transformation > Click '**Set Actor Rotation**' function.



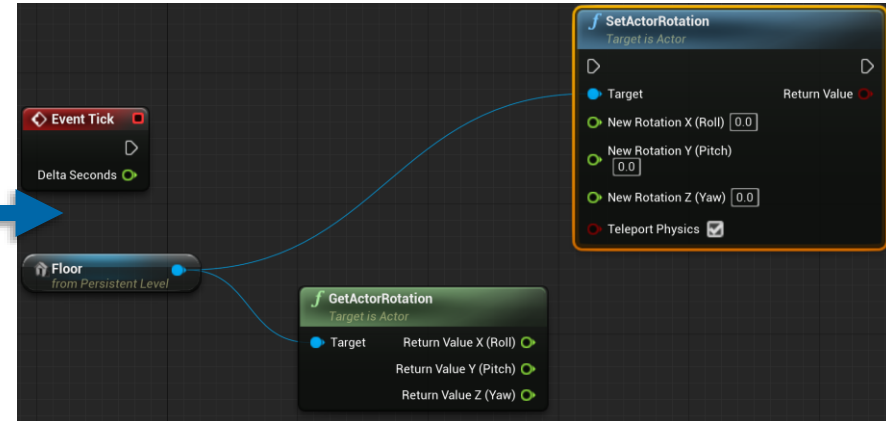
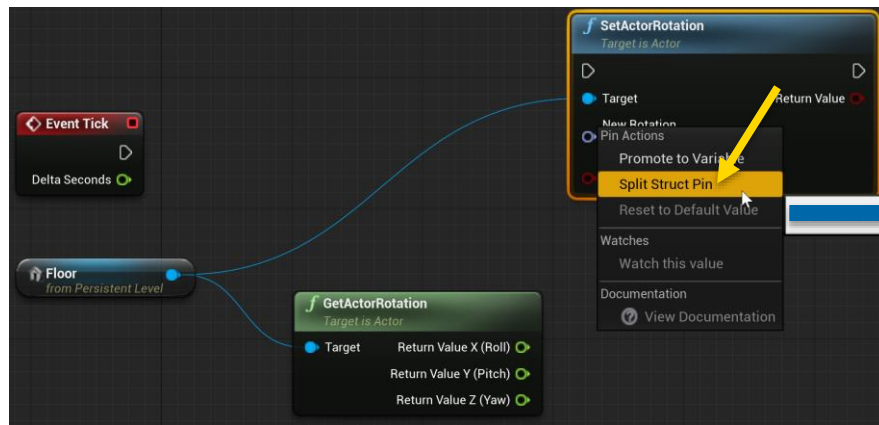
Tilt the floor with mouse input (Offset issue)

- Make the **Teleport Physics** active.



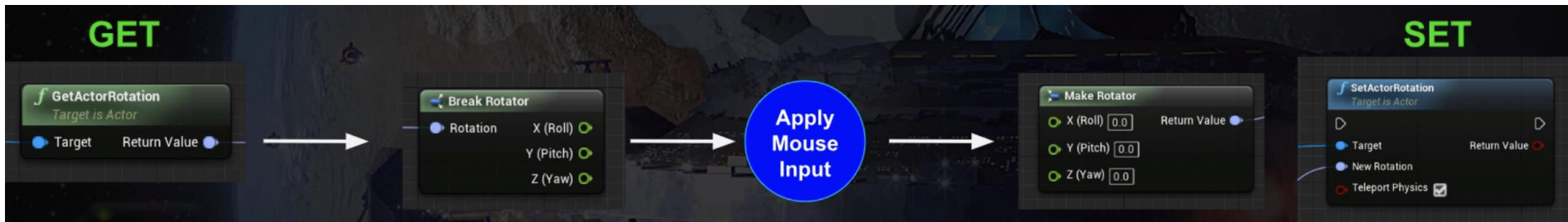
Tilt the floor with mouse input (Offset issue)

Split the **New Rotation** pin.



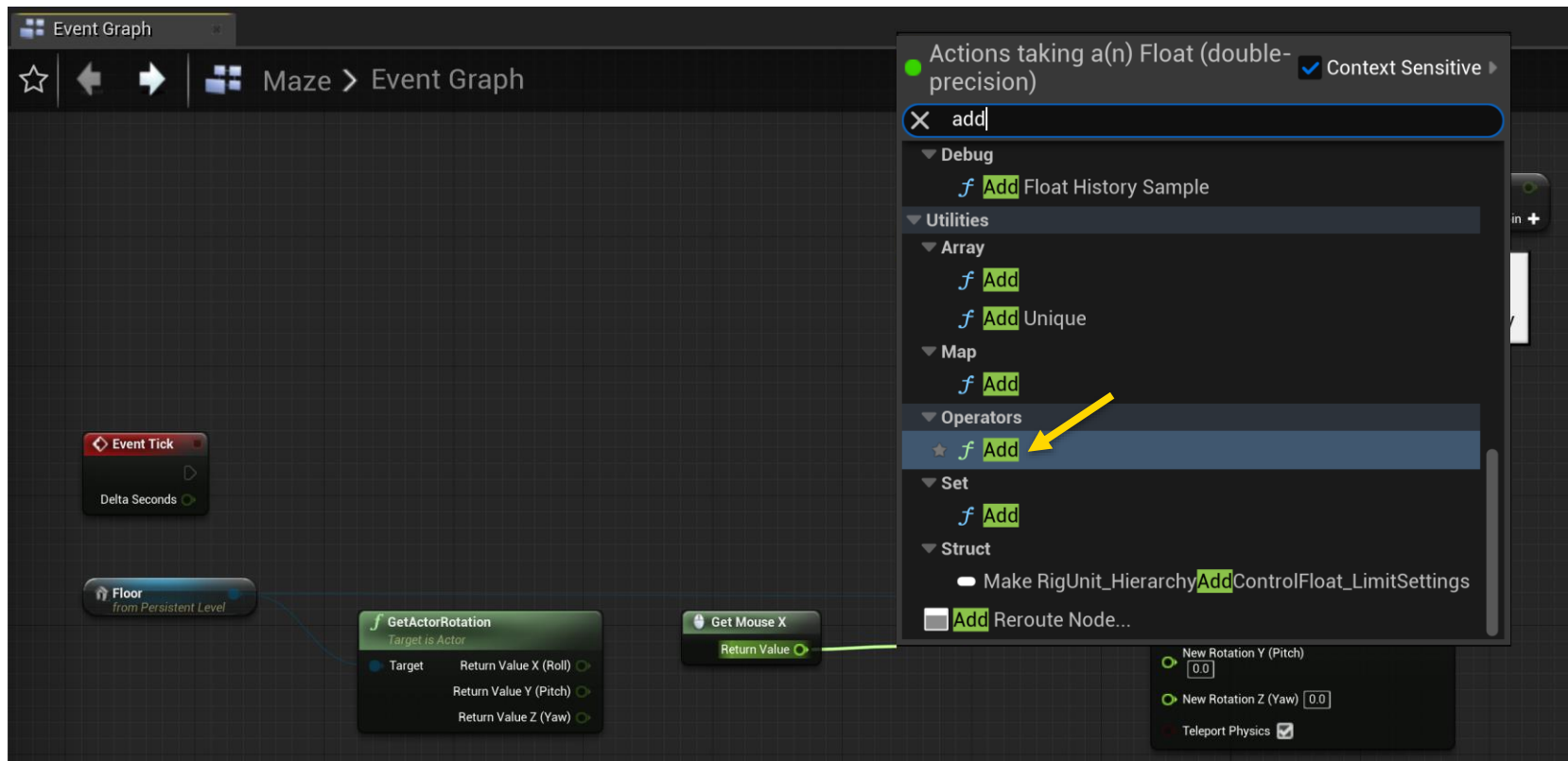
Tilt the floor with mouse input (Offset issue)

- **Get Actor Rotation** function, returns the current actors' rotation.
- **Set Actor Rotation** function sets a new rotation for the actor.
- We can apply our mouse input using some simple math calculation between these nodes.



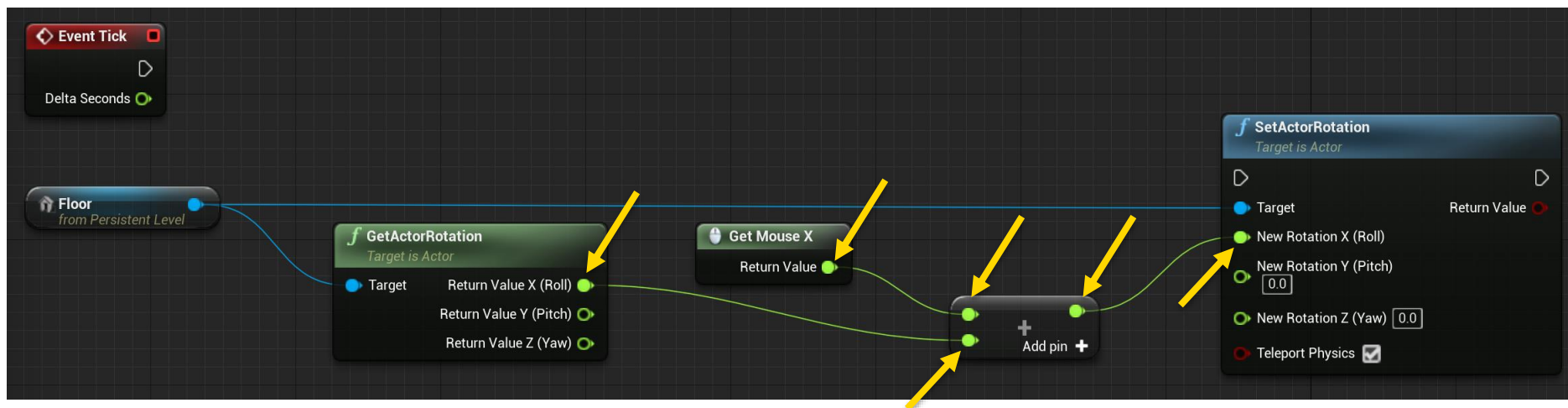
Tilt the floor with mouse input (Offset issue)

- Drag out from the 'Mouse X' node return value pin and find 'Add' function.



Tilt the floor with mouse input (Offset issue)

- Connect 'Get Mouse X' node and 'Get Actor Rotation' Return Value X (Roll) pin to the input pins of the 'Add'.
- Connect 'Add' output pin to New Rotation X (Roll) pin of the 'Set Actor Rotation' function.

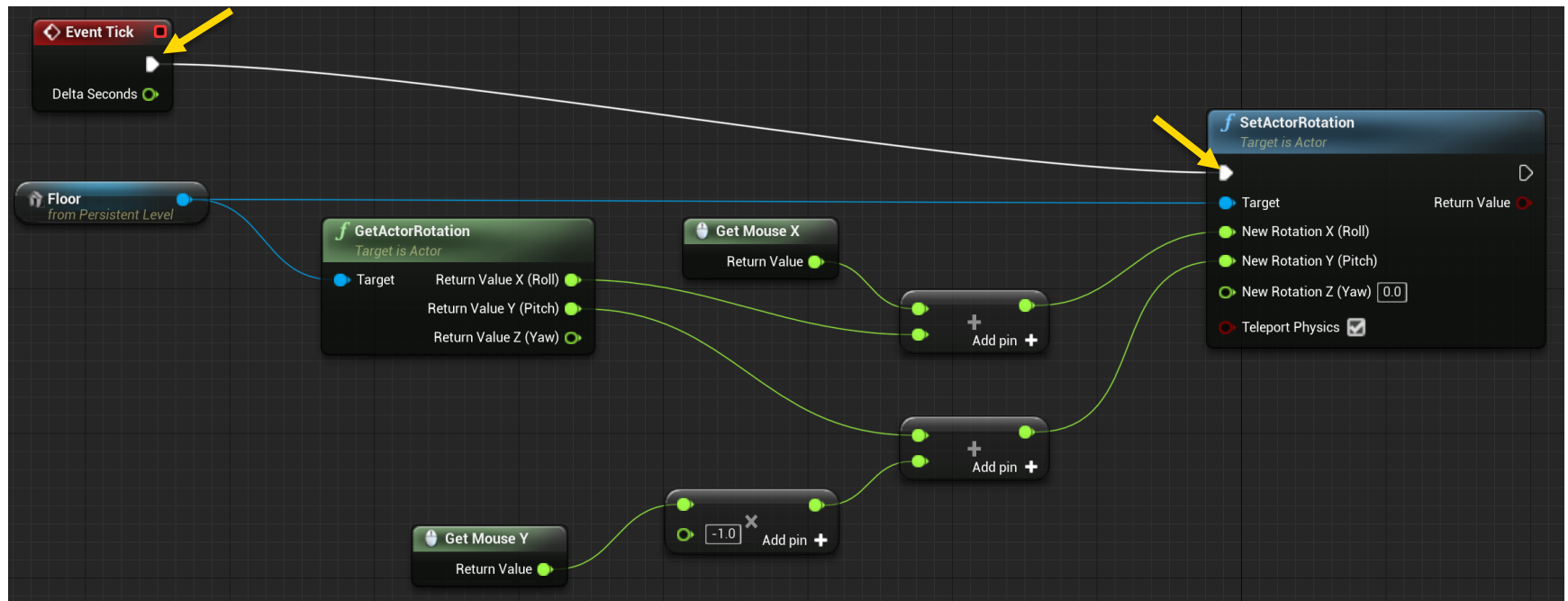


- Connect 'Get Mouse Y' and 'Get actor rotation' Return Value Y (Pitch) pin to the input pins of the 'Add'.
- Connect 'Add' output pin to New Rotation Y (Pitch) of the 'Set Actor Rotation' function.



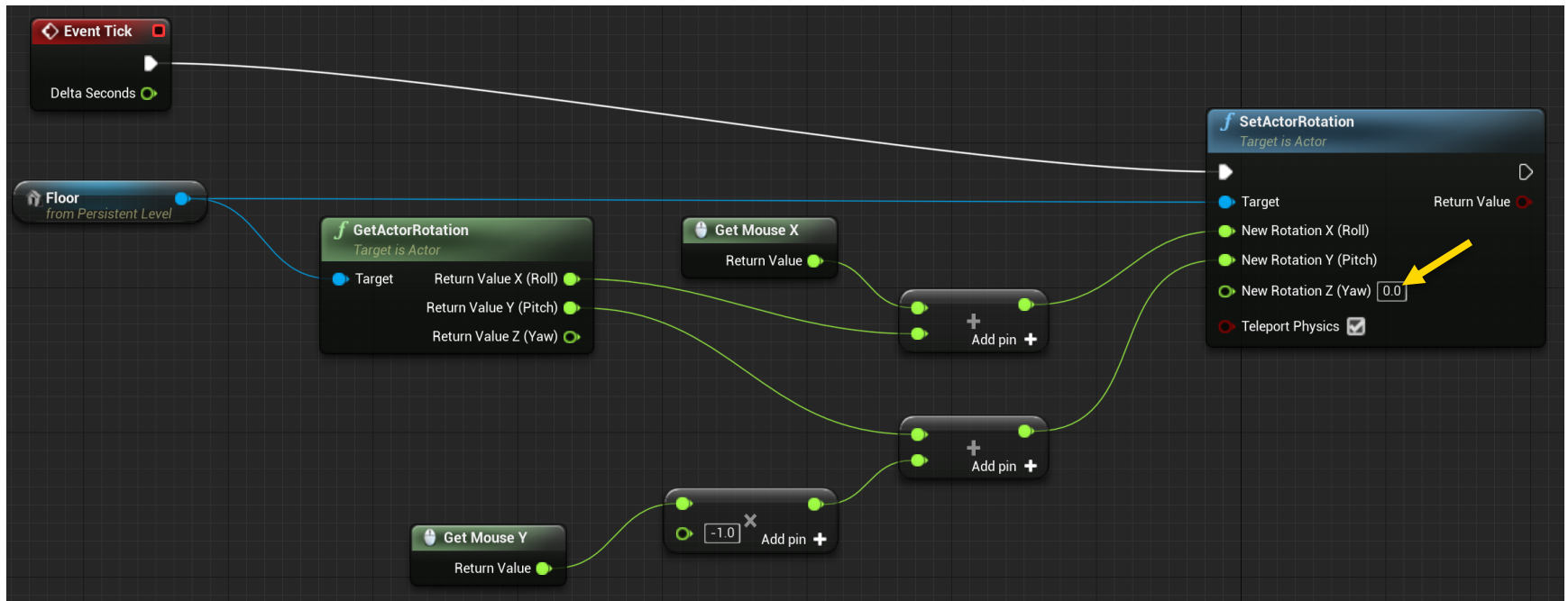
Tilt the floor with mouse input (Offset issue)

- Connect **Event Tick** to the input pin of the '**Set Actor Rotation**' function to make the code functional.

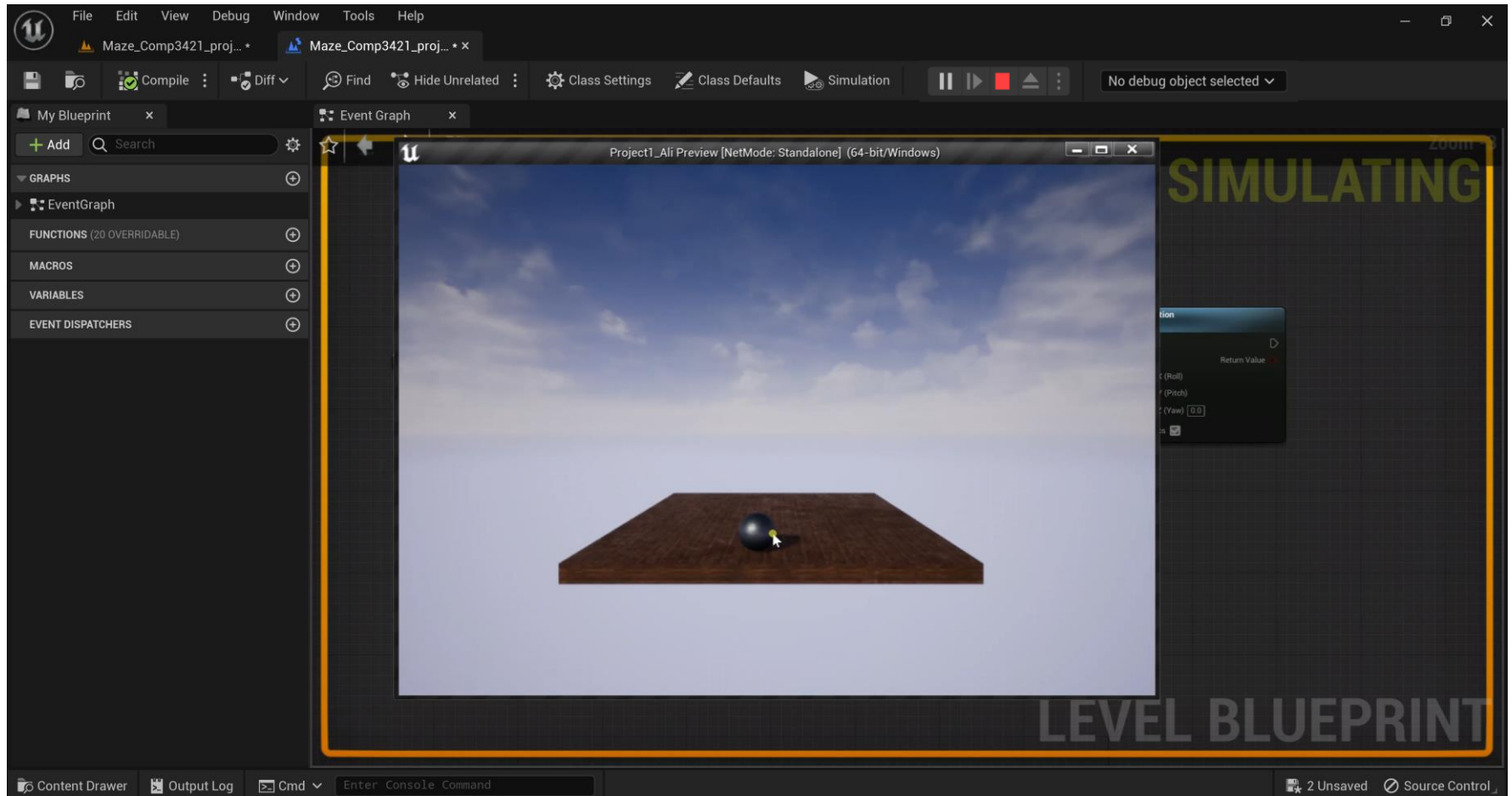


Tilt the floor with mouse input (Offset issue summary)

- ‘Set Actor Rotation’ function sets a new rotation for the floor by using ‘Get Actor Rotation’ function that returns the current floor’ rotation + the current mouse rotation ($\text{Get Mouse X} / \text{Get Mouse Y}$) → Floor rotation value.
- Always the z axis will be kept at 0.



Tilt the floor with mouse input (Offset issue)



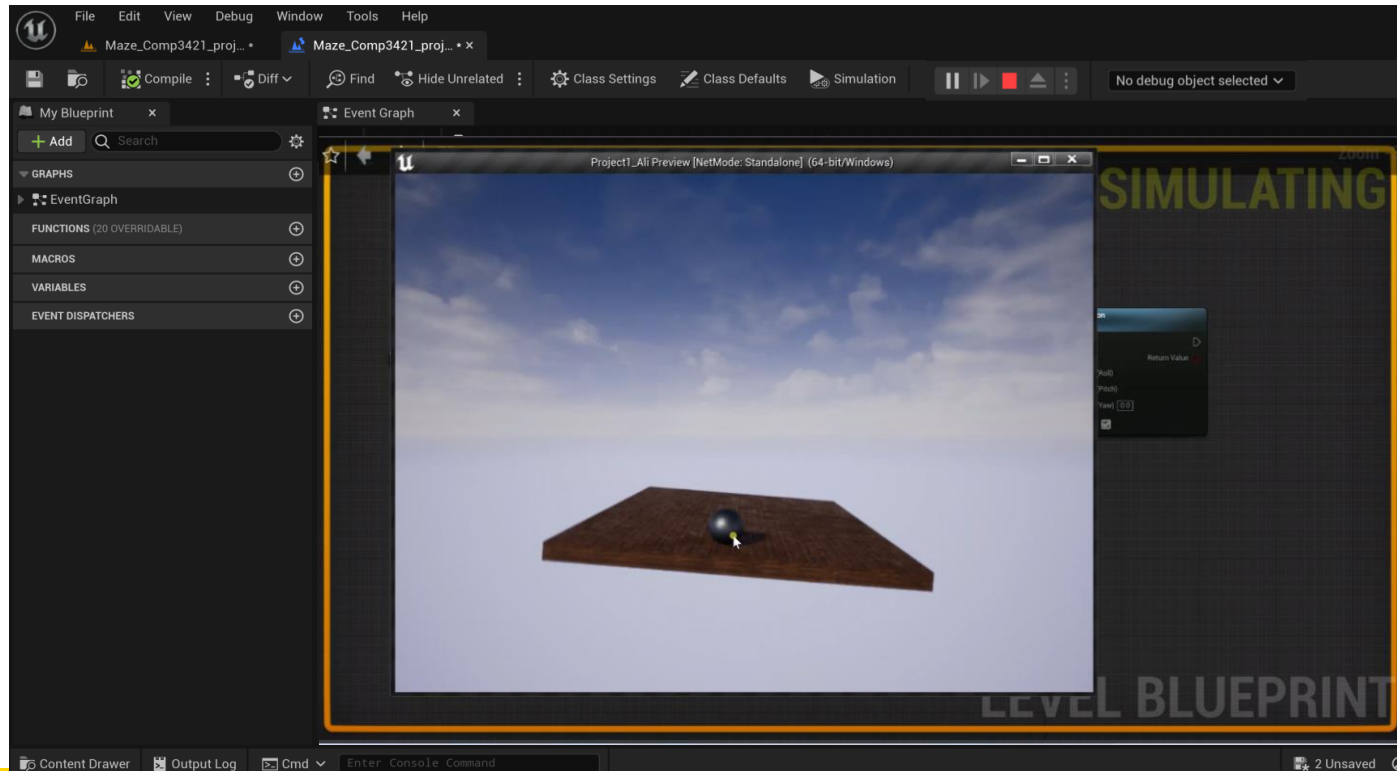
Lecture activity 5

Take the mouse X and Y values and add them to the current Roll and pitch rotation of the floor actor to solve the floor rotation offset issue by keeping the z axis = 0.



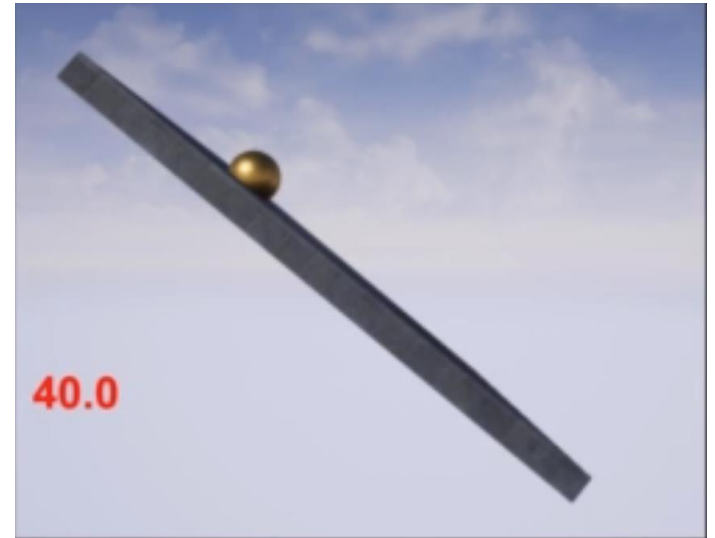
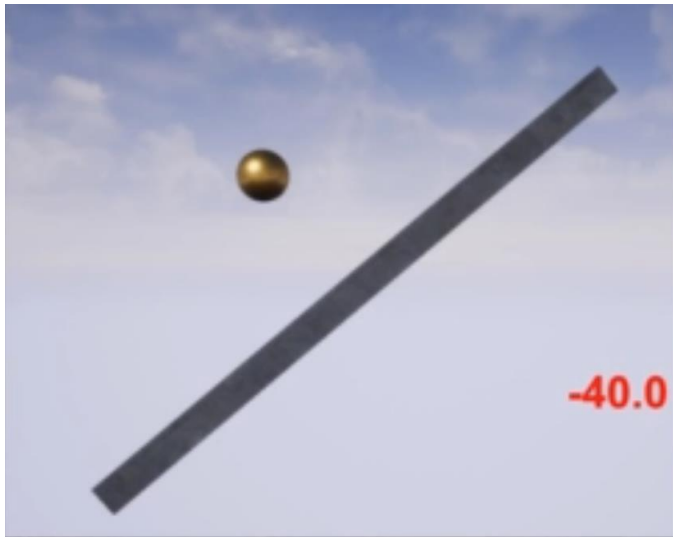
Limiting how far we can rotate the floor

When we move our mouse very fast in different axes, the floor will keep flipping all the way round.



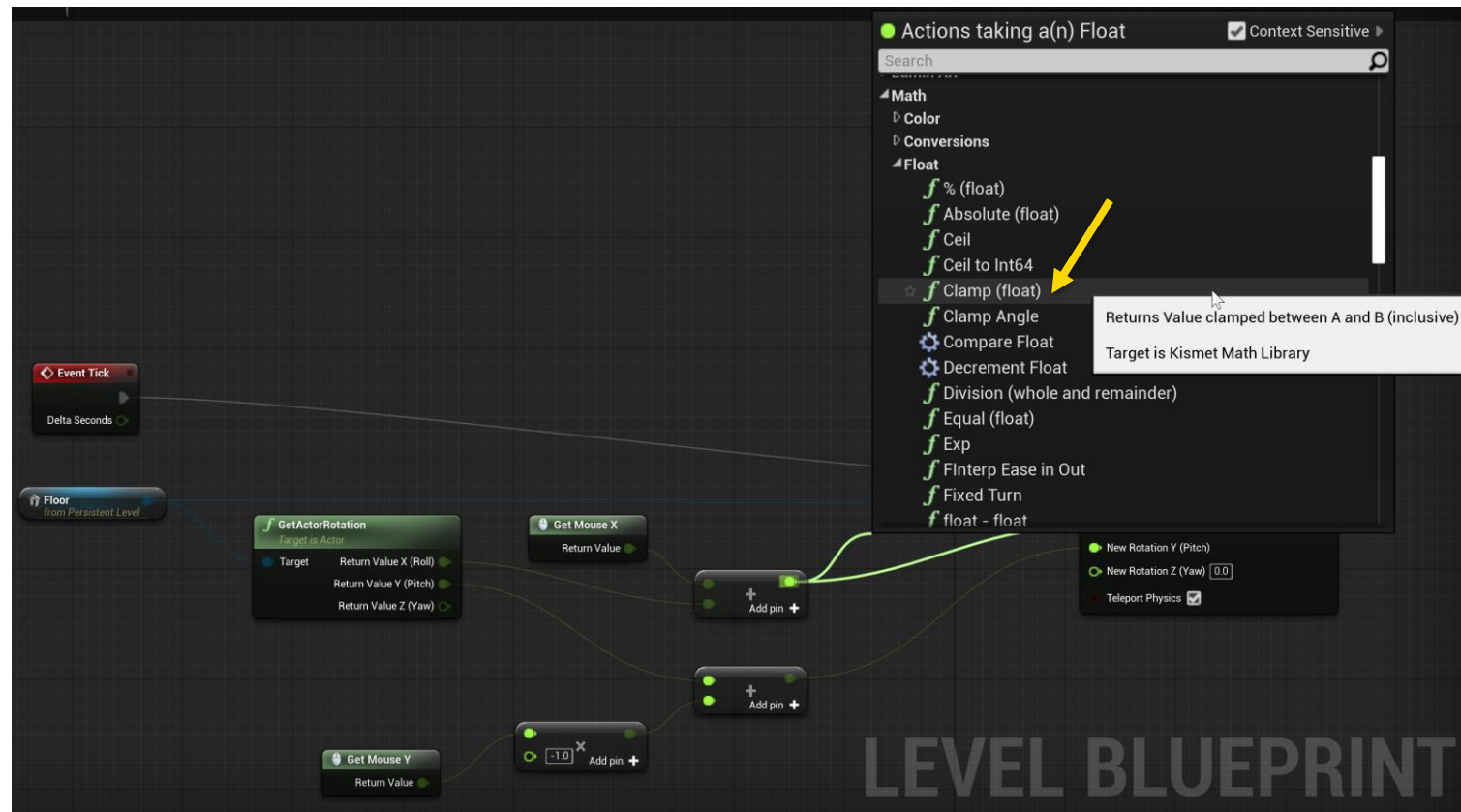
Limiting how far we can rotate the floor

Limit the floor movement range to - 40 and 40.



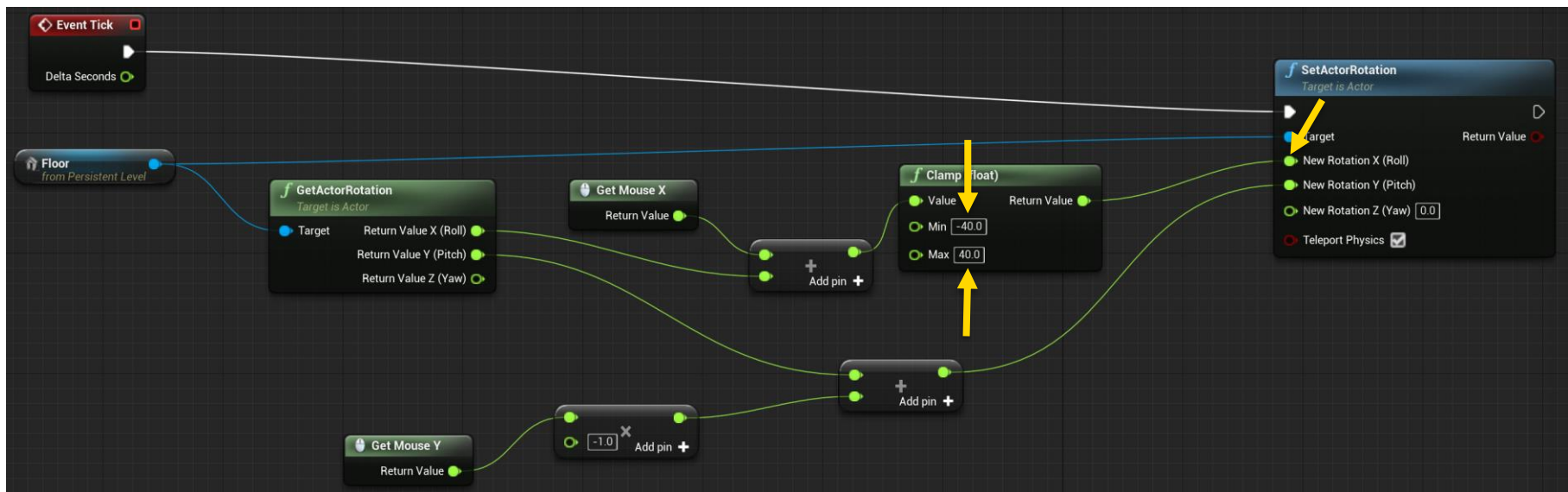
Limiting how far we can rotate the floor

Drag out the return value from the **Addition** node > Math category > Float section > **Clamp (float)** function.



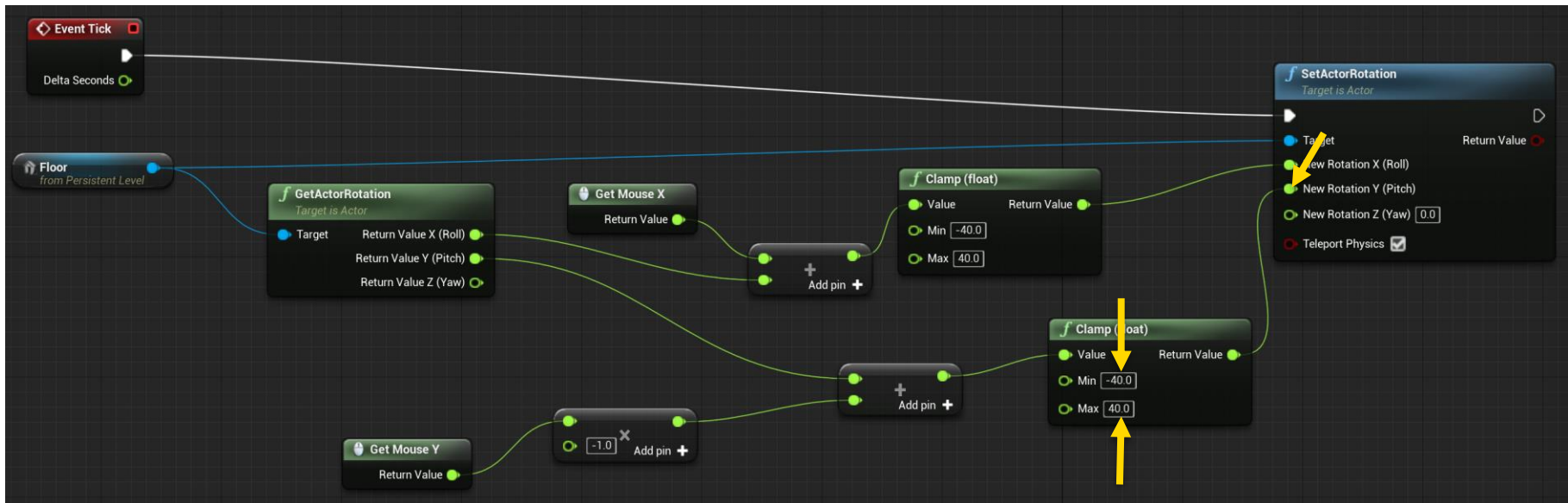
Limiting how far we can rotate the floor

- **Clamp (float)** function has a min and max value.
- Put -40 in Min and 40 in Max.
- Connect the return value into the **Roll** pin of the '**Set Actor Rotation**' function.

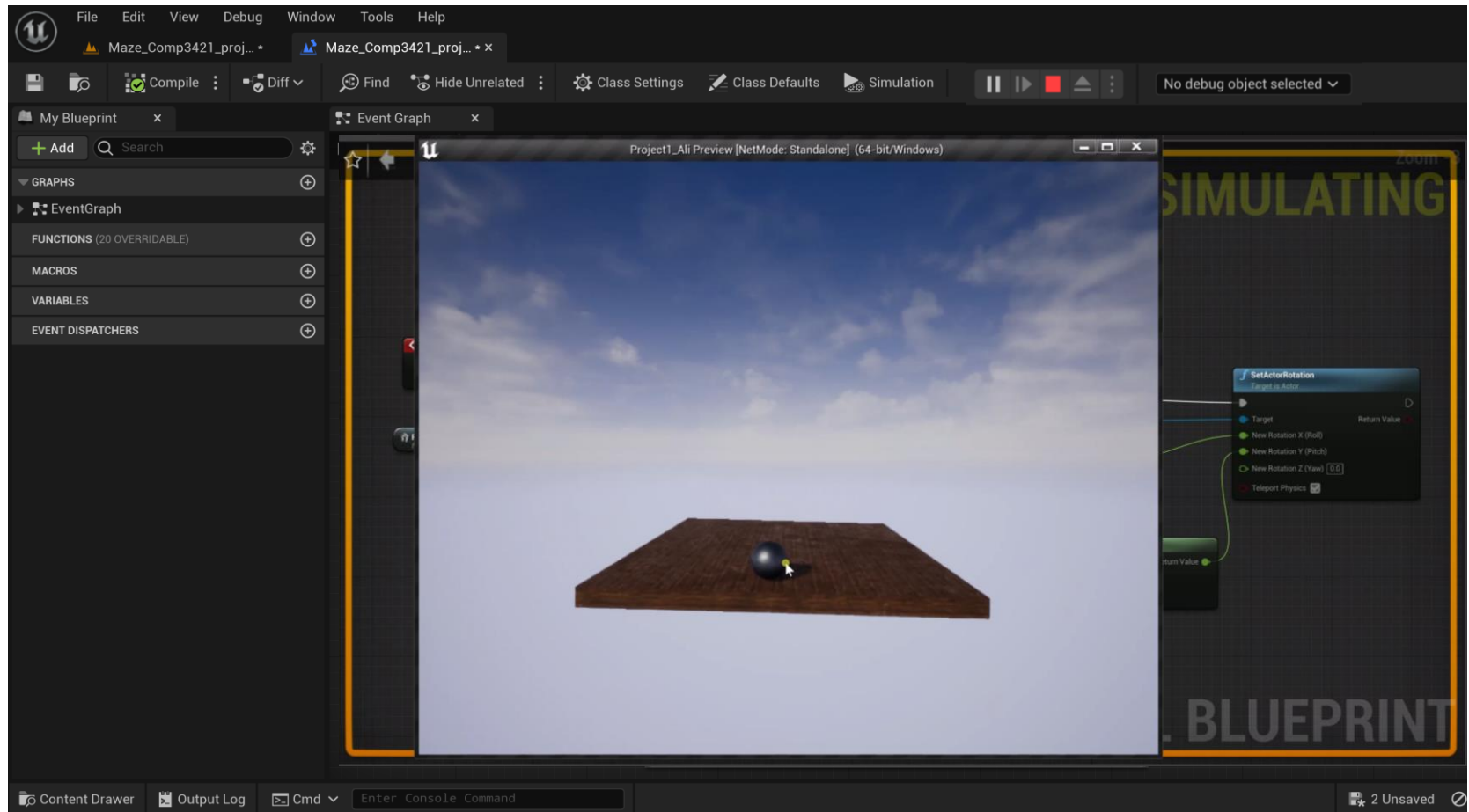


Limiting how far we can rotate the floor

- **Clamp (float)** function has a min and max value.
- Put -40 in Min and 40 in Max.
- Connect the return value into the **Pitch** pin of the '**Set Actor Rotation**' function.



Limiting how far we can rotate the floor



Lecture activity 6

Limiting how far we can rotate the floor by clamping the floor movement.



